RESEARCH ON FACTORS INFLUENCING CUSTOMER LOYALTY THROUGH SATISFACTION TOWARD MOBIFONE TELECOMMUNICATIONS CORPORATION IN HO CHI MINH CITY

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Từ khóa:
MobiFone; sự hài lòng; lòng trung thành; chất lượng dịch vụ; giá trị cảm nhận; chăm sóc khách hàng.

TÓM TẮT

Nghiên cứu xem xét vì sao khách hàng tiếp tục gắn bó với MobiFone khi chất lượng mạng, giá trị nhận được, chăm sóc khách hàng và trải nghiệm số ngày càng dễ được đặt cạnh các lựa chọn thay thế. Dữ liệu được thu thập từ 450 khách hàng tại TP.HCM và phân tích bằng Cronbach's Alpha, EFA, hồi quy tuyến tính, các kiểm định so sánh nhóm và đối chiếu 2.821 phản hồi của 50 chi nhánh Mobifone tại TP.HCM từ Google Maps. Kết quả cho thấy lòng trung thành không hình thành từ một điểm chạm đơn lẻ mà tích lũy qua cách khách hàng đánh giá toàn bộ trải nghiệm. Chất lượng dịch vụ cảm nhận và chất lượng chăm sóc khách hàng là hai tín hiệu gắn mạnh nhất với sự hài lòng; giá trị cảm nhận, tính dễ sử dụng và tính hữu ích tiếp tục bổ sung cho đánh giá đó. Khi nhìn toàn bộ chuỗi trải nghiệm → sự hài lòng → lòng trung thành, mức giải thích tổng hợp đạt 65,3%. Tỷ lệ này không có nghĩa mọi điểm chạm đều đã tốt: sự hài lòng và lòng trung thành đều vượt ngưỡng 4, trong khi năm yếu tố đầu vào vẫn thấp hơn ngưỡng này. Khoảng lệch cho thấy khách hàng có thể vẫn ở lại dù phải chịu những ma sát nhỏ lặp lại ở kết nối, hỗ trợ, giá trị nhận được và thao tác số; rủi ro giữ chân xuất hiện khi các ma sát đó làm việc chuyển một phần nhu cầu sang nhà mạng khác trở nên dễ hơn việc tiếp tục chịu bất tiện. Vì vậy, MobiFone không chỉ cần nâng chuẩn kỹ thuật mà còn phải làm cho từng lần sử dụng ổn định, dễ được hỗ trợ và đủ xứng đáng để khách hàng không cần tìm một lựa chọn khác.

Keywords:
MobiFone; customer satisfaction; customer loyalty; service quality; perceived value; customer care.

ABSTRACT

This study examines why customers continue to remain loyal to MobiFone as network quality, perceived value, customer care, and digital experience are increasingly compared with alternative providers. Data were collected from 450 customers in Ho Chi Minh City and analyzed using Cronbach's Alpha, exploratory factor analysis (EFA), linear regression, group comparison tests, and a cross-check against 2,821 Google Maps reviews covering 50 MobiFone branches in Ho Chi Minh City. The findings indicate that customer loyalty is not formed through a single touchpoint but accumulates through customers' evaluation of the overall experience. Perceived service quality and customer care quality are the two factors most strongly associated with satisfaction, while perceived value, perceived ease of use, and perceived usefulness further contribute to this evaluation. When the full pathway of experience → satisfaction → loyalty is considered, the model achieves an overall explanatory power of 65.3%. This figure does not imply that every touchpoint is already performing well: both satisfaction and loyalty exceed the threshold of 4, whereas the five antecedent factors remain below this level. This gap suggests that customers may continue to stay despite repeatedly encountering minor frictions in connectivity, support, perceived value, and digital interactions. Retention risk emerges when these frictions make shifting part of their usage to another mobile operator easier than continuing to tolerate the inconvenience. Therefore, MobiFone needs not only to raise its technical standards but also to ensure that every interaction is stable, easy to support, and sufficiently worthwhile so that customers have little reason to seek an alternative.

1. INTRODUCTION

By mid-2025, competition in Vietnam's mobile telecommunications market was no longer simply a race in SIM numbers. At the end of September 2024, the country had about 119,3 million mobile subscribers, down 4,8% year on year; by 10/10/2024, the number of 2G-only subscribers had fallen to 771.072, below 1% of total subscribers, before 2G service for 2G-only devices ceased on 15/10/2024 (General Statistics Office, 2024; Government Electronic Newspaper, 2024). Taken together, the two figures show that the market has moved beyond a growth stage driven mainly by subscriber numbers: the harder task is retaining users within a data and digital-service ecosystem in which experiences across operators are increasingly easy to compare.

For MobiFone, 2024 was also a period when traditional telecommunications services declined while the company had to invest in the transition from 4G to 5G; MobiFone announced that its 2024 pre-tax profit exceeded plan by 20,1% and that many digital services grew rapidly (MobiFone, 2024). On 27–28/02/2025, the right to represent the State's ownership of capital in MobiFone was transferred to the Ministry of Public Security; less than one month later, on 26/03/2025, MobiFone officially launched commercial 5G. After one month, the company stated that the 5G service had more than 2,5 million users, with Ho Chi Minh City among the prominent markets for usage (Government Electronic Newspaper, 2025; MobiFone, 2025a, 2025b). The expansion of network capacity therefore occurred alongside a more difficult requirement: turning technology investment into an experience customers can feel in every call, data session, package purchase or support request.

Higher technical capability does not automatically translate into loyalty. When speed and coverage improve, that new standard itself becomes the customer's new reference point; a dropped call, a prolonged support request or a package viewed as poor value will be compared with the best available alternative, not with the quality of several years ago. At the same time, mobile number portability lowers part of the barrier to leaving, while the use of multiple SIMs and digital platforms lowers comparison costs. The retention problem therefore lies in the gap between actual experience and rapidly rising expectations: customers do not need to reject the entire brand to reduce their attachment; they only need to shift part of their data, communication or digital-service needs to another provider when small frictions recur often enough.

Studies in Vietnam show that satisfaction often lies between service experience and loyalty, but each study explains only part of this chain. Trịnh Kim Hoa and Lưu Thị Bích Ngọc (2015) show that in Vietnam's mobile telecommunications sector, service quality, brand image and price perception are all positively associated with satisfaction, while satisfaction is positively associated with loyalty. Nguyễn Thị Quỳnh Nga and Lê Đặng Minh Thu (2019), studying mobile telecommunications customers in Ho Chi Minh City, further identify satisfaction with the service as the component most strongly associated with brand loyalty. Nguyễn Hải Quang (2024) extends the context to fiber-optic Internet in Ho Chi Minh City and shows that service quality is associated with both satisfaction and loyalty, with satisfaction serving as a partial bridge. Nguyễn Minh Trí, Từ Quang Phương and Thái Vân Hà (2024) add evidence from Viettel by placing technical quality, system quality, service quality and price quality alongside perceived value and satisfaction. The studies reviewed therefore clarify several individual links, but they have not directly placed MobiFone customers in Ho Chi Minh City within the 5G transition while simultaneously examining service quality, perceived value, customer care and digital-service usage experience in one unified structure. From this gap, the study “Research on factors influencing customer loyalty through satisfaction toward MobiFone Telecommunications Corporation in Ho Chi Minh City” was conducted to identify the associations between experience dimensions and satisfaction, clarify the association between satisfaction and loyalty, test evaluations against an expectation benchmark, differences by gender and usage duration, and compare bottlenecks with feedback across the store network. The ultimate objective is to identify the touchpoints that should be prioritized so that technical and service quality become a sufficiently clear reason for customers to stay.

2. RESEARCH METHODOLOGY

2.1 Theoretical basis and research model

Parasuraman and colleagues (1985) laid the foundation for service-quality research by viewing quality as the gap between expectations and actual experience; Parasuraman, Zeithaml and Berry (1988) later operationalized this approach through SERVQUAL. Zeithaml (1988) added the perceived-value perspective, emphasizing the balance between benefits received and costs incurred by customers. Davis (1989) extended the discussion to technology experience through Perceived Usefulness and Perceived Ease of Use, while Oliver (1999) placed loyalty at a later stage of the evaluation process, when accumulated experience is sufficient to form an intention to maintain the relationship.

In the Vietnamese context, Trịnh Kim Hoa and Lưu Thị Bích Ngọc (2015) show that service quality, brand image and price perception are positively associated with satisfaction, while satisfaction is positively associated with loyalty in mobile telecommunications. Nguyễn Thị Quỳnh Nga and Lê Đặng Minh Thu (2019), using a sample of mobile telecommunications customers in Ho Chi Minh City, further identify service satisfaction as the component most strongly associated with brand loyalty. Nguyễn Hải Quang (2024) extends the evidence to fiber-optic Internet service in Ho Chi Minh City by showing that service quality is associated with both satisfaction and loyalty, with satisfaction acting as a partial bridge.

From the above arguments, the proposed research model includes five factors associated with Customer Satisfaction (CS): Perceived Service Quality (PSQ), Perceived Value (PV), Customer Care Quality (CCQ), Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Satisfaction is placed before Customer Loyalty (CL) to test whether the overall evaluation of the experience is associated with the desire to continue the relationship; service usage duration is retained as a control variable to examine whether the main relationships remain stable when accumulated experience differs.

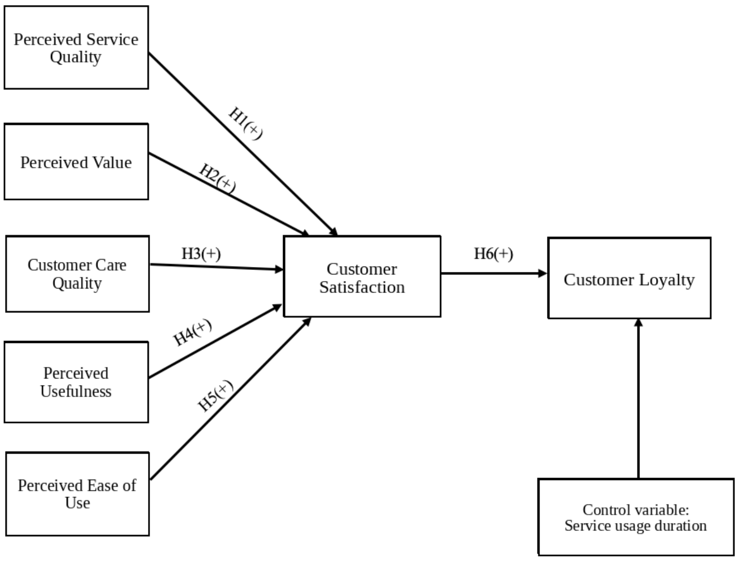


Figure 1: Proposed research model

2.2 Analysis method

Data analysis is designed sequentially from scale assessment to testing relationships among variables; this section defines only the processing approach and decision criteria, without interpreting results. The first step uses Cronbach's Alpha to assess the internal consistency of each scale. According to Nunnally (1978), Alpha of about 0,70 or higher is generally accepted for established scales; the study also uses a corrected item-total correlation above 0,30 as a condition for retaining an observed variable.

The second step uses exploratory factor analysis (EFA) with Principal Component Analysis and Varimax rotation. EFA is conducted separately for the independent-variable group and the CS-CL group to test whether the observed variables converge on the expected structure. The criteria for reading the results include KMO of at least 0,50, Bartlett Sig.<0,05, primary factor loading of at least 0,50 and total variance explained above 50% (Hair and colleagues, 2010). A variable is considered for removal only when it fails to achieve the primary loading or when cross-loading destroys the ability to distinguish among factors.

The third step uses multiple linear regression to estimate the associations of PSQ, PV, CCQ, PU and PEOU with CS, followed by simple linear regression to estimate the association of CS with CL. Standardized Beta is used to compare relative association strength among variables in the same equation; VIF checks multicollinearity and Durbin-Watson is used as a diagnostic indicator of residual autocorrelation. Because the data are collected at one point in time, regression coefficients are interpreted as conditional predictive relationships within the sample and are not automatically converted into causal conclusions.

Three supplementary tests answer questions different from regression. A One-Sample T-Test uses the Likert benchmark of 4 to determine whether the mean evaluation differs from "Agree"; an Independent Samples T-Test compares the two gender groups; ANOVA compares usage-duration groups and uses Tamhane when necessary to identify the group pairs responsible for differences. Finally, Google Maps reviews are used as a field cross-check layer to identify touchpoints that commonly appear in operational feedback. Google Maps data do not replace the survey, and topic frequency in reviews is not interpreted as a statistical effect coefficient.

2.3 Data collection method

The survey respondents are individual customers aged 18 or older who primarily live or work in Ho Chi Minh City and have used MobiFone services for at least six months. The preliminary study discussed the wording with 10 customers; the official quantitative survey was conducted in 07-08/2025 using convenience sampling and obtained 450 valid responses. The questionnaire uses 37 observed variables across seven scales—PSQ, PV, CCQ, PU, PEOU, CS and CL—measured on a 5-point Likert scale from 1 "Strongly disagree" to 5 "Strongly agree."

Sample size was determined before analysis to ensure sufficient observations for EFA and regression. With 37 observed variables, 450 responses correspond to about 12,2 observations per variable. Hair and colleagues (2010) consider 5:1 a commonly used minimum ratio and 10:1 a more conservative level in factor-analysis practice; therefore, a sample of 450 meets the technical requirements of the model. The limitation lies in the convenience-sampling method rather than the number of observations: the descriptive results reflect the study sample, not an absolute probability estimate for all MobiFone customers in Ho Chi Minh City.

3. RESULTS AND DISCUSSION

3.1 Survey sample characteristics

Table 1: Characteristics of the MobiFone customer survey sample

Characteristics	Sample n = 450		
	Frequency	Percentage %	Cumulative %
Gender			
Male	226	50.2	50.2
Female	224	49.8	100
Age			
From 18 to 25	134	29.8	29.8
From 26 to 35	130	28.9	58.7
From 36 to 50	136	30.2	88.9
Over 50	50	11.1	100
Occupation			
Student	104	23.1	23.1
Office employee	96	21.3	44.4
Teacher/Researcher	71	15.8	60.2
Worker/general laborer	79	17.6	77.8
Business owner	77	17.1	94.9
Other	23	5.1	100
Marital status			
Single	277	61.6	61.6
Married	173	38.4	100
Income			
Under 5 million	115	25.6	25.6
From 5 to under 10 million	124	27.6	53.1
From 10 to under 20 million	136	30.2	83.3
20 million or more	75	16.7	100
Education			
High school or vocational secondary school	95	21.1	21.1
College	83	18.4	39.6
University	176	39.1	78.7
Postgraduate	73	16.2	94.9
Other	23	5.1	100
Product/service used			
Voice service (calls, text messages)	82	18.2	18.2
Data service (4G/5G)	139	30.9	49.1
Digital services (My MobiFone app, digital content)	90	20.0	69.1
All of the above	139	30.9	100
Product/service usage duration			
From 6 months to under 1 year	109	24.2	24.2
From 1 year to under 3 years	109	24.2	48.4
From 3 years to under 5 years	115	25.6	74.0
5 years or more	117	26.0	100
Purpose of use			
Personal communication	183	40.7	40.7
Work/study	136	30.2	70.9
Entertainment (internet access, watching videos, gaming)	115	25.6	96.4
Other	16	3.6	100
Average spending			
Under VND 100,000	117	26.0	26.0
From VND 100,000 to under VND 200,000	142	31.6	57.6
From VND 200,000 to under VND 500,000	135	30.0	87.6
VND 500,000 or more	56	12.4	100
Previously used product/service			
Used before	318	70.7	70.7
Never used	132	29.3	100

Source: Descriptive statistics from the study data, n=450, 2025.

The three age groups from 18 to 50 account for 88,9% of the sample, while data services or integrated service packages appear among 61,8% of respondents. This structure places connection quality and digital experience where they are most likely to be experienced frequently. However, the sample proportions do not by themselves prove that this is the structure of the entire market because convenience sampling was used; the value of the table is to show who is generating the coefficients in the subsequent sections.

Two sample characteristics are directly relevant to the loyalty analysis. A total of 70,7% of respondents have used another operator, showing that most customers have experience comparing providers. The four usage-duration groups range from 24,2% to 26,0%, limiting the possibility that one tenure group dominates the ANOVA results. However, cross-sectional data reflect only associations at the survey time and cannot identify the temporal direction between usage tenure and loyalty; it is therefore considered as a control variable in the model.

3.2 Results of scale reliability testing

All seven scales have Cronbach's Alpha above 0,70: PSQ = 0,939; PV = 0,846; CCQ = 0,831; PU = 0,859; PEOU = 0,884; CS = 0,784 and CL = 0,748. The lowest corrected item-total correlation is 0,505 for CL3 and the highest is 0,780 for PSQ7, both above the 0,30 threshold. Therefore, no observed variable is removed before EFA.

Table 2: Scale reliability test results

Alpha	Variable	Item-total c orrelation	Alpha if i tem deleted
PSQ (0.939)	PSQ1	0,751	0,933
	PSQ2	0,742	0,933
	PSQ3	0,742	0,933
	PSQ4	0,766	0,932
	PSQ5	0,758	0,933
	PSQ6	0,756	0,933
	PSQ7	0,780	0,932
	PSQ8	0,745	0,933
	PSQ9	0,773	0,932
	PSQ10	0,714	0,935
PV (0.846)	PV1	0,669	0,810
	PV2	0,663	0,813
	PV3	0,729	0,784
	PV4	0,669	0,810
CCQ (0.831)	CCQ1	0,681	0,776
	CCQ2	0,634	0,797
	CCQ3	0,656	0,788
	CCQ4	0,664	0,784
PU (0.859)	PU1	0,676	0,830
	PU2	0,702	0,823
	PU3	0,675	0,830
	PU4	0,650	0,836
	PU5	0,674	0,830
PEOU (0.884)	PEOU1	0,751	0,853
	PEOU2	0,711	0,862
	PEOU3	0,684	0,868

Alpha	Variable	Item-total c orrelation	Alpha if i tem deleted
	PEOU4	0,737	0,856
	PEOU5	0,725	0,859
CS (0.784)	CS1	0,548	0,748
	CS2	0,574	0,739
	CS3	0,579	0,737
	CS4	0,559	0,744
	CS5	0,539	0,751
CL (0.748)	CL1	0,548	0,686
	CL2	0,569	0,674
	CL3	0,505	0,710
	CL4	0,547	0,687

Source: Cronbach Alpha test results from the study data, 2025

PSQ has the highest Alpha but still does not exceed the range that commonly suggests excessive content repetition; at the same time, all ten PSQ variables have item-total correlations from 0,714 to 0,780. Because PSQ covers the network, responsiveness, employees and transaction points, all ten variables moving consistently together is a necessary condition for the later PSQ regression coefficient to remain meaningful. Reliability answers only “do the items move together?”; it does not answer “are the groups truly distinct from one another?”, so EFA remains a necessary step.

3.3 Factor analysis results

EFA of the 28 input variables yields KMO = 0,896; Bartlett χ^2 = 6.596,854, df = 378 and p<0,001. Five factors with Eigenvalue greater than 1 explain 66,376% of total variance. No variable is removed because all primary loadings exceed 0,50. The rotated matrix preserves the five structures PSQ, PEOU, PU, PV and CCQ rather than creating new groups different from the theoretical model.

The two pairs of concepts requiring closer examination are PEOU-PU and PV-CCQ. PEOU loads from 0,797 to 0,850 while PU loads from 0,778 to 0,819; PV loads from 0,814 to 0,857 while CCQ loads from 0,795 to 0,834. This separation allows a distinction between a service that is “easy to operate” and one that is “actually useful,” and between the feeling of “value for money” and the experience of “being well supported.” If these pairs were mixed together, the Beta ranking in regression would be difficult to translate into specific managerial decisions.

Table 3: Rotated factor matrix analysis results

Variable	F1	F2	F3	F4	F5
PSQ7	**0,828**	0,013	0,005	-0,011	-0,011
PSQ9	**0,822**	-0,021	0,038	0,035	0,019
PSQ4	**0,815**	0,000	-0,029	-0,034	0,003
PSQ5	**0,809**	0,001	-0,015	-0,020	-0,005
PSQ6	**0,806**	0,008	-0,047	0,010	-0,063
PSQ1	**0,802**	0,046	-0,015	0,020	-0,016
PSQ8	**0,797**	-0,040	0,006	-0,033	0,013
PSQ3	**0,795**	-0,072	0,031	0,013	0,026
PSQ2	**0,795**	0,029	-0,007	0,058	0,001
PSQ10	**0,770**	0,038	-0,018	-0,054	-0,004
PEOU1	0,032	**0,850**	0,001	-0,024	0,052
PEOU4	-0,046	**0,839**	0,059	0,023	0,046
PEOU5	-0,010	**0,828**	-0,012	0,037	-0,001
PEOU2	0,061	**0,817**	0,056	0,007	-0,025
PEOU3	-0,032	**0,797**	-0,033	0,043	0,001
PU2	0,025	-0,018	**0,819**	0,053	0,017
PU1	-0,060	-0,006	**0,800**	0,010	-0,005
PU3	-0,040	0,053	**0,799**	0,002	-0,018
PU5	0,030	0,045	**0,798**	0,025	-0,012
PU4	0,006	-0,008	**0,778**	0,035	0,001
PV3	0,036	0,001	0,024	**0,857**	0,072
PV4	-0,016	0,048	0,008	**0,815**	0,066
PV2	-0,053	0,015	0,006	**0,815**	-0,005
PV1	0,052	0,016	0,084	**0,814**	-0,040
CCQ1	0,014	-0,008	0,003	-0,053	**0,834**
CCQ4	-0,027	0,065	-0,027	-0,016	**0,818**
CCQ3	0,006	0,081	0,000	0,074	**0,807**
CCQ2	-0,016	-0,072	0,006	0,083	**0,795**
Eigen.	6,490	3,564	3,212	2,792	2,528
% Var.	23,171	12,326	11,482	9,856	9,540
Alpha	0,939	0,884	0,859	0,846	0,831

KMO = 0,896; Sig. Bartlett < 0,001; Cumulative = 66,376%

EFA for CS and CL

Variable	F1: CS	F2: CL
CS4	**0,743**	0,141
CS3	**0,732**	0,180
CS2	**0,704**	0,240
CS1	**0,701**	0,197
CS5	**0,592**	0,376
CL2	0,161	**0,777**
CL4	0,150	**0,762**
CL1	0,281	**0,688**
CL3	0,268	**0,651**

KMO = 0,888; Sig. Bartlett < 0,001; Cumulative = 55,517%

Source: EFA factor analysis results from the study data, 2025.
Source: EFA factor analysis results from the study data, 2025.

EFA of the 9 CS-CL variables continues to separate into two factors: the five CS variables converge on one factor and the four CL variables on the other; KMO = 0,888 and total variance explained = 55,517%. Because satisfaction and loyalty are not merged into one measurement block, the CS -> CL regression in the next step has meaning as a relationship between two different constructs rather than between two different names for the same scale.

3.4 Linear regression analysis results

Multiple linear regression with CS as the dependent variable yields R = 0,704, R^2 = 0,496 and adjusted R^2 = 0,490; F = 87,419 with p <

0,001. Thus, PSQ, PV, CCQ, PU and PEOU together explain 49,6% of the variance in Satisfaction in the sample. Durbin-Watson = 2,166 and the largest VIF = 1,010 do not indicate meaningful autocorrelation or multicollinearity. By standardized Beta, the associations decrease in the order PSQ (0,389), CCQ (0,336), PV (0,298), PEOU (0,260) and PU (0,206). This ranking answers which variable carries more unique predictive information when all five factors appear in the same equation; it does not turn Beta into a budget-contribution ratio or causal evidence.

In the next equation, simple linear regression CS → CL yields R = 0,557, R^2 = 0,311, adjusted R^2 = 0,309, F = 202,001 and p < 0,001; Durbin-Watson = 1,951. The coefficient B = 0,530 indicates that when CS increases by 1 point, CL is predicted to increase by 0,530 points in the linear model; standardized Beta = 0,557 shows that this is the strongest association among the estimated regression paths. However, R^2 = 0,311 also establishes a clear limit: CS alone explains only 31,1% of CL variance, so loyalty also depends on factors not included in this equation.

Table 4: Linear regression analysis results

Variable name	B	Beta	Sig.	VIF
CS model				
Constant	-0,638	-	0,007	-
PSQ	0,333	0,389	<0,001	1,000
PV	0,250	0,298	<0,001	1,010
CCQ	0,286	0,336	<0,001	1,005
PU	0,190	0,206	<0,001	1,006
PEOU	0,221	0,260	<0,001	1,004
CL model				
Constant	1,937	-	<0,001	-
CS	0,530	0,557	<0,001	1,000

CS = -0,638 + 0,333PSQ + 0,250PV + 0,286CCQ + 0,190PU + 0,221PEOU

CL = 1,937 + 0,530CS

Source: Linear regression analysis results from the study data, 2025.

The combined explanatory level of the two-equation chain is calculated using the R^2m index: $R^2m = 1 - (1 - R_1^2)(1 - R_2^2) = 1 - (1 - 0,496)(1 - 0,311) = 0,6527 \approx 0,653$. Here, R_1^2 = 0,496 reflects the portion of Satisfaction variance explained by PSQ, PV, CCQ, PU and PEOU; R_2^2 = 0,311 reflects the portion of Loyalty variance explained by Satisfaction. R^2m = 0,653 shows that the two-equation chain reaches a combined explanatory level of 65,3%. This index reflects the explanatory power of the full research structure and does not replace R^2 for each equation; confirming mediation effects along each path requires additional estimation of indirect effects and bootstrap confidence intervals.

3.5 One-Sample T-Test results

With a test benchmark of 4, all five input variables have Mean below 4; the largest gap is PV (-0,3044), followed by PSQ (-0,2853), PEOU (-0,2604), PU (-0,2120) and CCQ (-0,0906). Conversely, CS = 4,1880 and CL = 4,1583 are both above 4 with p<0,001. The difference is not contradictory: CS and CL are overall evaluations, whereas the five input factors require respondents to look at specific touchpoints; the overall experience can be good while some individual frictions remain.

Table 5: One-Sample T-Test results for the scales

Variable	Mean	SD	t	Sig.	Difference from 4
PSQ	3,7147	0,78167	-7,743	<0,001	-0,28533
PV	3,6956	0,79554	-8,118	<0,001	-0,30444
CCQ	3,9094	0,78473	-2,448	0,015	-0,09056
PU	3,7880	0,72286	-6,221	<0,001	-0,21200
PEOU	3,7396	0,78680	-7,022	<0,001	-0,26044
CS	4,1880	0,66777	5,972	<0,001	0,18800
CL	4,1583	0,63541	5,286	<0,001	0,15833

Source: One-Sample T-Test results from the study data, 2025.

3.6 Independent Samples T-Test results

By gender, PSQ, PV, PU, PEOU and CL do not differ significantly at the 5% level. Two differences appear in CCQ and CS: women rate CCQ higher than men (4,0703 versus 3,7500; p<0,001) and CS higher than men (4,2741 versus 4,1027; p=0,006). CL is nearly unchanged between the two groups (4,1518 versus 4,1648; p=0,828). Gender is therefore not a strong enough signal to infer a difference in loyalty; the observed difference lies in customer-care experience and satisfaction.

Table 6: T-Test results comparing variables by gender

Variable	Male(n=226)	Female(n=224)	Sig.	Conclusion
PSQ	3,7398	3,6893	0,493	No difference
PV	3,6726	3,7188	0,539	No difference
CCQ	3,7500	4,0703	<0,001	Different
PU	3,7973	3,7786	0,783	No difference
PEOU	3,7425	3,7366	0,937	No difference
CS	4,1027	4,2741	0,006	Different
CL	4,1648	4,1518	0,828	No difference

Source: Independent Samples T-Test results from the study data, 2025.

3.7 ANOVA results for the control variable of usage duration

ANOVA uses the four usage-duration levels only as a supplementary comparison for the control variable. PSQ, PV, CCQ, PU and PEOU do not differ significantly among the four groups, while CS and CL both have p<0,001. CL rises from 3,7798 in the 6-<12 month group to 4,0665; 4,3022 and 4,4551 in the three longer-duration groups. CS is also higher among longer-term users but does not increase completely linearly. The cross-sectional data therefore leave two mechanisms open: relationship accumulation over time and a survivor effect, in which less-compatible customers leave earlier and therefore no longer appear in the high-tenure groups.

Table 7: ANOVA and Tamhane comparisons by usage duration

Variable	6-<12m	1-<3y	3-<5y	>=5y	Sig.	Conclusion
PSQ	3,7459	3,7229	3,7313	3,6615	0,855	No difference
PV	3,7638	3,7431	3,7478	3,5363	0,094	No difference
CCQ	3,8670	3,9450	3,9543	3,8718	0,757	No difference
PU	3,8183	3,7835	3,6922	3,8581	0,346	No difference
PEOU	3,7028	3,8661	3,7026	3,6923	0,293	No difference
CS	3,9413	4,2477	4,1826	4,3675	<0,001	Different
CL	3,7798	4,0665	4,3022	4,4551	<0,001	Different
Tamhane - significantly different pairs						
CS	6-<12m ↔ 1-<3y				0,001	Difference
CS	6-<12m ↔ 3-<5y				0,049	Difference
CS	6-<12m ↔ >=5y				<0,001	Difference
CL	6-<12m ↔ 1-<3y				0,004	Difference
CL	6-<12m ↔ 3-<5y				<0,001	Difference
CL	6-<12m ↔ >=5y				<0,001	Difference
CL	1-<3y ↔ 3-<5y				0,023	Difference
CL	1-<3y ↔ >=5y				<0,001	Difference

Source: ANOVA and Tamhane test results from the study data, 2025.

3.8 Field cross-check: high overall scores do not erase operational frictions

To examine whether bottlenecks appear outside the questionnaire, 2.821 Google Maps reviews for 50 MobiFone stores in Ho Chi Minh City were cross-checked for the period 15/07/2025-15/07/2026; 49/50 stores had reviews during the period. Across the entire dataset, 1.872 reviews contain text and 949 contain star ratings only; the average rating is 4,306/5. This score indicates a fairly positive overall image, but it does not answer the more important operational question: when the experience breaks down, where are customers getting stuck?

Table 8: Cross-check between quantitative survey results and Google Maps feedback traces

Evidence layer	Result	Significance
Field scope	50 stores; 49/50 have reviews in the period	Broad cross-check across the Ho Chi Minh City store network, not a single transaction point.
Feedback volume	2.821 reviews; 1.872 with text; 949 star-only	Content analysis should rely only on reviews with text; star-only reviews are still retained to calculate the score distribution.
Overall score	4,306/5; 2.229 five-star reviews	A high average score describes the general state and does not rule out recurring failure modes.
Problem reviews	418 reviews with 1-3 stars and text	This is a diagnostic set of pain points, not a representation of the share of dissatisfied customers in the entire market.
Prominent content groups	CCQ 278/418 (66,5%); PSQ 191/418 (45,7%); PEOU 109/418 (26,1%); CS->CL 14/418 (3,3%)	Groups may overlap; mention frequency reflects review content, not a statistical effect coefficient.

Source: Google Maps data for 50 MobiFone stores in Ho Chi Minh City, period 15/07/2025-15/07/2026.

The cross-check focuses on the two load-bearing areas PSQ and CCQ, but feedback frequency on Google Maps need not match the Beta ranking in regression. PSQ's Beta of 0,389 reflects the association across the full experience of 450 customers; store-level Google Maps reviews capture more situations in which customers must transact, seek support or resolve problems. Therefore, CCQ appearing in 278/418 problem reviews (66,5%) while PSQ ranks first in the regression is not a contradiction: one side shows which factor is more strongly associated with satisfaction, while the other shows which failures are most visible when customers need to visit a service point. Channel bias must remain an open consideration because store reviews can make customer-care problems appear more frequently than they do across the full usage journey.

The 4,306/5 score on Google Maps also does not negate the 418 problem reviews. Most reviews are 5-star ratings, while the pain-point diagnosis focuses on negative reviews with text; the two statistics answer different questions. The value of this data layer lies in checking the diagnostic direction and identifying specific operational failures, not in turning Google Maps into a test that replaces the survey. The cross-check dataset, store list and Review IDs can be viewed here.

3.9 Synthesis of research model testing results

All six H1-H6 paths have the expected sign and are statistically significant. The first model explains 49,6% of CS variance; the second explains 31,1% of CL variance, with CS having Beta = 0,557. When the two equations are combined, $R^2m = 0,653$ shows that the chain has an overall explanatory level of 65,3%. Usage duration continues to be retained in its proper role as a control variable directed toward CL; ANOVA is only a supplementary comparison among usage-duration levels and does not become a grouping variable in the theoretical model.

Table 9: Summary of research model testing results

Hypothesis	Relationship and result	Conclusion
H1	PSQ -> CS; Beta=0,389; p<0,001	Accepted
H2	PV -> CS; Beta=0,298; p<0,001	Accepted
H3	CCQ -> CS; Beta=0,336; p<0,001	Accepted
H4	PU -> CS; Beta=0,206; p<0,001	Accepted
H5	PEOU -> CS; Beta=0,260; p<0,001	Accepted
H6	CS -> CL; Beta=0,557; p<0,001	Accepted
Control variable	Usage duration -> CL; control variable; supplementary ANOVA p<0,001	Accepted
Model	$R^2m = 0,653$; combined $R^2(CS)=0,496$ and $R^2(CL)=0,311$	Combined explanatory level

Source: Testing results from the study data, 2025.

The Beta ranking places PSQ ahead of CCQ, PV, PEOU and PU in the CS equation; H6 has Beta = 0,557, the largest among the estimated regression paths. $R^2m = 0,653$ adds a chain-level interpretation, but it does not replace a mediation test: two consecutive regressions and a combined index do not show how large the individual indirect effect of PSQ, PV, CCQ, PU or PEOU through CS is. Therefore, the most defensible conclusion is that the five factors are associated with CS and CS is strongly associated with CL; the degree of mediation for each path requires bootstrap testing if it is to be asserted directly.

4. CONCLUSION AND MANAGERIAL IMPLICATIONS

4.1 Conclusion

The results show that MobiFone customer loyalty is formed through a chain of accumulated experiences rather than a single touchpoint. The five input factors together explain 49,6% of the variance in Satisfaction; standardized Beta decreases from Perceived Service Quality (0,389), Customer Care Quality (0,336), Perceived Value (0,298), Perceived Ease of Use (0,260) to Perceived Usefulness (0,206). Satisfaction is in turn positively associated with Loyalty ($\beta = 0,557$; $R^2 = 0,311$). When the two equations are viewed as a chain, the combined explanatory index R^2m reaches 65,3%. The 65,3% figure indicates the general explanatory scope of the research structure but does not replace an indirect-effect test; with cross-sectional data, the appropriate conclusion remains that the variables move together according to the expected structure rather than that causal relationships have been established.

The gap between overall evaluations and specific touchpoints is the part that matters most managerially. Satisfaction reaches 4,1880 and Loyalty 4,1583, both above the benchmark of 4, while all five input factors remain below this benchmark. Customers may therefore still be satisfied with MobiFone even when some component experiences do not meet the corresponding expectation level. A cross-check of 2.821 Google Maps reviews across 50 stores shows an average score of 4,306/5 and 2.229 five-star reviews; however, among the 418 one-to-three-star reviews containing text, Customer Care Quality appears 278 times (66,5%), Service Quality 191 times (45,7%) and Ease of Use 109 times (26,1%). A high overall score therefore does not erase recurring frictions when customers need support, encounter technical problems or must complete a digital task on their own.

Retention risk is also not evenly distributed across customer groups. Women rate Customer Care Quality and Satisfaction higher than men, but Loyalty does not differ significantly by gender; therefore, a better customer-care experience cannot be inferred to create a different loyalty mechanism between the two groups. By usage duration, the 6-to-under-12-month group has the lowest Satisfaction (3,9413) and the lowest Loyalty (3,7798). The early stage of the customer relationship is therefore more vulnerable: a poor experience occurring before usage habits and switching costs have had time to form is more likely to lead customers to try an alternative. This result must be understood within the convenience sample in Ho Chi Minh City, not as the churn rate of all MobiFone customers.

Customers do not wake up every morning asking themselves whether they are loyal to MobiFone. They simply notice whether calls are stable, whether data works when needed, whether someone follows a problem through to resolution, whether the package is worth the money paid and whether the application forces them to spend extra effort. Each good experience is only a small plus, but a chain of recurring frictions gradually makes staying more costly in time, patience and inconvenience. The largest risk therefore is not necessarily that customers immediately discard their SIM; they may keep the number while quietly shifting data, calls or digital services to another operator. Loyalty is then lost before the subscriber disappears from the system. MobiFone truly retains customers only when technical capability is translated into a stable everyday usage experience that is easy to support and worthwhile enough that users do not need to seek another option.

4.2 Managerial implications in priority order

The solution system centers on the Satisfaction-Loyalty link. CS has a standardized Beta of 0,557, the highest among the estimated regression paths; CS alone explains 31,1% of CL variance, while the combined explanatory level of the research chain reaches $R^2m = 0,653$. Because Satisfaction is formed from preceding experiences, improvement resources should focus on the five input factors in descending effect order: PSQ, CCQ, PV, PEOU and PU. Mean scores identify experience dimensions that remain low, Beta coefficients indicate the relative association between each factor group and CS, and Google Maps feedback adds field signals about operational bottlenecks. The three sources of evidence are combined to determine priority order and intervention content.

Priority 1 — Improve Perceived Service Quality (PSQ, β = 0,389)

PSQ has both the highest Beta and several touchpoints below the benchmark of 4: rapid response to requests reaches 3,66; services matching individual needs 3,69; coverage 3,70 and internet speed 3,77. Three actions need to be implemented in parallel. First, organize multi-tier support: repetitive queries move to self-service, while complex problems need clear SLAs and correct routing from the first contact; SLAs should be set separately by channel and request complexity, then monitored using the median and processing-time percentiles rather than a single average. Second, use usage behavior to personalize packages and recommendations rather than sending the same offer to every customer. Third, make coverage and the network-upgrade roadmap, especially 5G, transparent so customers know what quality is already available and which areas are still being improved. PSQ is not only an infrastructure issue; it is the sum of technical quality, response speed and need fit, so it cannot be fixed with one KPI.

Priority 2 — Optimize customer care (CCQ, β = 0,336)

CCQ reaches 3,9094 and is a relative strength, but complaint-resolution speed (CCQ2) reaches only 3,88; men rate CCQ at 3,7500, lower than women's 4,0703. The objective is therefore not to open as many channels as possible, but to make existing channels solve the problem. The causes of men's lower ratings should first be studied in depth rather than assuming gender itself creates the difference. Operationally, frontline staff need enough authority to increase First Contact Resolution, while CRM 360° should let them see interaction history so customers do not have to repeat the same problem across multiple touchpoints. Support channels that are already rated well should be maintained; additional resources should focus on diagnostic capability and end-to-end resolution. The fact that CCQ appears in 278/418 problem reviews on Google Maps clarifies this gap: customers may be able to reach MobiFone, but what they remember is whether the problem was resolved.

Priority 3 — Strengthen Perceived Value (PV, β = 0,298)

PV has Mean 3,6956 and PV4—"competitive value compared with other operators"—reaches only 3,66. This gap does not simply mean that tariffs are too high; it shows that customers do not yet see a sufficiently clear advantage when comparing MobiFone with alternatives. Communication should therefore shift from "how much it costs, how much data you get" to the total value received: data allowance, digital services, partner benefits and benefits that can be used immediately. The Kết Nối Dài Lâu program should also link usage tenure and spending with tangible benefits such as priority support, data or vouchers. If prices are merely cut across the board, customers may pay less but PV4 may still not rise because the core issue is perceived differentiation, not only absolute price.

Priority 4 — Improve Perceived Ease of Use (PEOU, β = 0,260)

PEOU reaches 3,7396; the item on a user-friendly interface reaches 3,71. My MobiFone should therefore be reviewed starting with the most frequent tasks such as checking the account, purchasing packages and contacting support. The objective of UI/UX is not to make the interface "prettier," but to reduce the number of steps users must remember, the number of choices they must consider and the probability of abandonment during the process. Key tasks should aim for completion in one or two taps when feasible, with recognizable icons, concise language and stable layouts. Each removed step must make the task clearer rather than merely making the screen emptier.

Priority 5 — Increase the usefulness of the digital ecosystem (PU, β = 0,206)

PU has the lowest Beta among the five factors, but Mean 3,7880 remains below the benchmark of 4. The digital ecosystem may therefore expand beyond account management through utilities genuinely tied to daily needs, such as bill payment, ticket booking or top-ups for digital services. However, adding a function creates value only when customers use it repeatedly. If a new utility does not increase usage frequency, shorten the journey or replace a task for which customers currently have to switch to another application, it only makes the interface heavier and directly harms PEOU. Usefulness and ease of use must be managed as a trade-off pair, not as two independent feature lists.

Cross-cutting solution — Care for the new-customer stage

Customers with less than one year of usage are the segment requiring the earliest intervention because CS is only 3,94 and CL 3,78. A structured six-month "Welcome Journey" should begin immediately after activation: brief guidance on services and the application, proactive contact after the initial period to detect difficulties, and a sufficiently clear starter benefit so customers can perceive value immediately. The important point is not giving more promotions but the timing of friction resolution. A technical error or an incomplete support experience in the first weeks has little positive history to offset it; early intervention therefore has more value than compensatory care after customers have already formed an intention to leave.

4.3 Boundaries of the conclusions

The conclusions are limited to the scope of the survey data. Convenience sampling reduces the ability to generalize absolute proportions to all MobiFone customers; the Ho Chi Minh City scope also does not fully reflect differences in provinces and rural areas. The cross-sectional design records variables at one point in time, so regression identifies predictive associations in the sample but does not establish temporal causal order. $R^2m = 0{,}653$ is a combined index of two equations, not the proportion of CL variance explained by a single variable and not a direct mediation-effect test. Beyond the measured variables, loyalty may also relate to brand image, social influence, switching costs and the attractiveness of alternatives. Because the survey uses self-reported attitudes, loyalty intention may also differ from actual churn behavior.

The Google Maps cross-check helps determine whether pain points appear outside the questionnaire but does not remove these limitations. Reviewers self-select, while the store channel is biased toward moments when customers need support; this data layer is therefore more suitable for identifying failure modes than estimating churn. Stronger evidence would link satisfaction scores, tickets, usage, renewals and port-out for the same customers over time. If PSQ or CCQ are truly the correct levers, improvements in corresponding operational indicators should appear first and move together with changes in satisfaction and service-retention behavior.

4.4 Post-implementation validation direction

Managerial priorities have value only when they produce observable changes in behavior and operations. For PSQ, monitoring should connect survey scores with technical traces such as data speed, failed-call rate, coverage and response time; for CCQ, processing time, first-contact resolution rate and the number of times customers must re-contact should be tracked. If satisfaction rises while operational indicators remain unchanged, the company should consider whether the change comes from sample composition or short-term expectations rather than real service improvement.

A stronger validation approach is to follow the same customer group over time. New customers can have CS and CL measured after the first month, third month and sixth month while complaints, usage and mobile-number-porting behavior are recorded. This design directly addresses the blind spot of the control variable: if the same individual increases CL as usage duration grows, the evidence leans toward relationship accumulation; if low-scoring customers leave the sample while high-scoring customers remain, the survivor effect provides a better explanation.

An investment decision should also be evaluated using before-after differences with a control group where conditions permit. For example, if SLA is shortened at a group of transaction points, the company can compare changes in CCQ, CS, re-contact rate and churn with a group that has not yet adopted the change during the same period. This approach is harder than reading Beta, but it answers the correct managerial question: does changing the process create the outcome or merely move together with it? When survey, operational and behavioral data all shift in the same direction, confidence in the decision increases.

4.5 Principles for translating results into managerial decisions

A large coefficient is useful only when it points to a place where intervention is possible. PSQ has the highest Beta, but PSQ is a block of ten touchpoints ranging from call and internet quality to responsiveness, employee attitude and facilities. Therefore, the decision to "invest in PSQ" remains too broad. Management must go one layer deeper: combine the block-level Beta with the Mean of each item and the corresponding operational trace. PSQ4 has Mean 3,66 while PSQ is the block most strongly associated with CS; response speed is therefore a point that should be examined first. Conversely, an item with a low Mean but belonging to a lower-Beta block is not automatically prioritized merely because it has the lowest score.

Quantitative results and field feedback complement one another at three levels. Mean scores show which aspects of the experience are rated low; Beta coefficients identify which factor groups are more strongly associated with Satisfaction; observed variables and Google Maps feedback point to specific touchpoints that should be prioritized for correction. Combining the three levels helps avoid concentrating resources on a single indicator and connects statistical results with operational problems at transaction points.

A good priority therefore needs to pass three gates at the same time. The first gate is association strength: the variable must have a sufficiently large Beta in the model under consideration. The second is the implementation gap: the item score or operating indicator must show room for improvement. The third is actionability: the company must have a specific process, system or decision capable of changing that touchpoint. Without the third gate, the statistical finding remains correct but is not yet a managerial project. With only the third gate and without the first two, a project can become an easy-to-do improvement rather than an improvement in the right place.

CS = 4,188 and CL = 4,158 create another perception risk: high overall scores can lead the organization to conclude that the experience is already fine. But all five input variables have Mean below 4, meaning the overall evaluation is higher than many specific components. This gap may come from relationship history, familiarity or benefits not fully captured in each scale; the current data do not allow one mechanism to be selected. Precisely because the durability of that buffer is unknown, the company should not use high CS and CL to delay resolving recurring frictions in PSQ and CCQ.

The control variable should be managed according to the logic of control, not as a lever that can be directly "increased." Usage duration is an accumulated outcome of the relationship between the customer and the operator; the company cannot order a new customer to become a five-year customer. The value of the Control Variable lies in holding tenure constant or tracking it to avoid confusing the influence of relationship history with the influence of current experience factors. ANOVA shows that CS and CL differ across usage-duration levels, but cross-sectional data do not show whether tenure makes customers more loyal or whether less-loyal customers have already left earlier.

Therefore, a program for new customers should be treated as a test of the mechanism rather than merely a care campaign. If customers with less than one year of usage have the lowest CS and CL, an early intervention must first change the frictions that this group encounters, and only then should CS and CL be expected to rise. If only survey scores increase while complaints, re-contact counts or churn behavior do not change, the company should suspect a measurement effect or changed expectations rather than quickly concluding that the program has created loyalty.

An appropriate monitoring system should connect three layers of data on the same timeline: self-reported customer experience, operational indicators at touchpoints and actual behavior. The first layer answers how customers feel; the second answers what the system did; the third answers what customers did afterward. The three layers do not necessarily move at the same time. Operational indicators often change first, perceptions may react after several experiences, and retention or churn behavior has a longer lag. Separating the three layers avoids requiring a single indicator to prove the entire story.

When resources are finite, the final principle is to prioritize where the cost of recurring friction exceeds the cost of fixing it. A rare but severe error and a mild but frequent error can create equivalent cumulative losses; the current questionnaire shows only perceptions and associations, not the cost of each failure mode. The next analysis cycle should therefore add incident frequency, processing time and post-incident behavior. Priority order would then no longer be based on the feeling of "which sounds more important," but on the combination of statistical effect, friction scale and intervention feasibility.

4.6 Design of the next measurement cycle

The next measurement cycle should begin with what the current model is missing: the study has identified experience blocks associated with CS and the CS-CL association, but it has not directly observed the process by which a customer moves from a specific incident to a behavioral change. To fill that gap, the unit of analysis must shift from "a survey response" to "a customer journey." Each CS or CL response should be linked, within permitted scope and with appropriate anonymization, to network-quality history, support requests, packages, usage levels and service-retention status. When the data layers lie on the same timeline, the company can determine whether a change in perception occurs before or after an operational change.

PSQ should be separated into indicators that the operations team can actually control. Call quality, data speed, coverage, response time and request-resolution capability are not the same type of failure even though they belong to one scale. If PSQ falls in an area, the first step is not to conclude that "service quality has declined," but to identify which component changed and whether it overlaps with where customers are complaining. This separation keeps the research construct from becoming an overly broad label while making the PSQ Beta executable rather than merely useful for ranking.

CCQ needs a similar logic, but the focus is on resolution outcomes. The number of contact channels or ease of access reflects only the entry point; poor experience is often determined by waiting time, how many times customers must repeat the problem and whether the request is resolved completely. Therefore, the next dataset should distinguish "received" from "resolved" and track repeat contacts for the same cause. If CCQ rises at the same time as first-contact resolution rises and repeat contacts fall, the chain of evidence will be much stronger than merely observing a higher survey score.

PV should be examined relative to alternatives rather than only the absolute bill. PV4 is the lowest item in the perceived-value group, showing that the weakness lies in competitive comparison; however, the current survey does not show which operator, package or benefit customers are comparing MobiFone against. A short follow-up question after the survey could record the alternative customers actually consider and why they see it as better value. Perceived value would then be converted from a general score into a specific competitive gap, helping avoid the default reaction of cutting prices when the real cause may lie in data allowance, coverage, digital services or package flexibility.

PEOU and PU should be measured through digital-platform behavior in parallel with the survey. An interface rated “easy to use” but with a high abandonment rate on an important task still contains friction that the overall scale does not see; conversely, a function rated useful but rarely used again may be attractive only in concept. Behavioral data such as task-completion rate, actual number of steps, repeat-use frequency and exit points in the process will show where customers struggle. The objective is not to collect as many logs as possible, but to link each log to a specific managerial question so that data do not become a storage repository that produces no decisions.

APPENDIX 1: QUANTITATIVE RESEARCH QUESTIONNAIRE

SURVEY ON SERVICE EXPERIENCE AT MOBIFONE

Dear Sir/Madam,

*I am currently a student at Nguyễn Tất Thành University. I am conducting the research topic "Resea**rch on factors influencing customer loyalty through the mediating role of customer satisfaction toward MobiFone Telecommunications Corporation in Ho Chi** Minh City." I would greatly appreciate your support by answering the questions in this survey.*

Please answer all questions; there are no right or wrong answers. Eligible survey participants are customers aged 18 or older who have used MobiFone services in Ho Chi Minh City for at least 6 months. The collected data are used only for academic research purposes and not for commercial purposes. I undertake that all information will be presented in aggregate statistical form and will not mention the name of any specific individual or organization. Thank you very much for your cooperation!

PART I. Please provide the following information:

To ensure the appropriateness of the survey, please answer several screening questions below.

Screening question 1: Do you currently live or work primarily in Ho Chi Minh City?

- ☐ Yes *(Please continue with the following questions)*
- ☐ No (Un*fortunately, this survey is not suitable for you. Thank you for your interest!)*

Screening question 2: As of now, have you used MobiFone mobile services for at least 6 months?

- ☐ Have used for 6 months or more (*Please continue with the following questions)*
- ☐ Less than 6 month*s (Unfortunately, this survey is not suitable for you. Thank you for your interest!)*

1. Which MobiFone service have you used? (Select 1)

☐ 1. Voice service (calls, text messages)
☐ 2. Data service (4G/5G)
☐ 3. Digital services (My MobiFone application, digital content)
☐ 4. All of the above

2. How long have you used MobiFone services? (Select 1)

☐ 1. From 6 months to under 1 year
☐ 2. From 1 year to under 3 years
☐ 3. From 3 years to under 5 years
☐ 4. 5 years or more

3. What do you primarily use MobiFone services for? (Select 1)

☐ 1. Personal communication
☐ 2. Work/study
☐ 3. Entertainment (internet access, watching videos, gaming)

☐ 4. Other (please specify): ………………

4. **On average, how much do you spend on MobiFone services each month? (Select 1)**

☐ 1. Under VND 100.000
☐ 2. From VND 100.000 to under VND 200.000
☐ 3. From VND 200.000 to under VND 500.000
☐ 4. VND 500.000 or more

PART II. Please indicate how you feel about MobiFone services according to the statements below.

We would like to hear your perceptions of your experience using MobiFone services. Please take a few minutes to evaluate the criteria below according to how well they match your perceptions.

For each statement, please mark X in one box from 1 to 5. By convention, the larger the number, the more you agree, specifically as follows:

Strongly disagree	Disagree	Mean	Agree	Strongly agree
1	2	3	4	5

No.	Statements	Level of agreement				
		Strongly disagree	Disagree	Mean	Agree	Strongly agree
1. Perceived Service Quality (PSQ)						
PSQ1	MobiFone's call quality is always stable.					
PSQ2	MobiFone's internet speed is reliable.					
PSQ3	MobiFone's coverage is extensive.					
PSQ4	MobiFone responds to my requests quickly.					
PSQ5	When I need support, MobiFone is always ready to help.					
PSQ6	MobiFone employees have good professional knowledge.					
PSQ7	MobiFone employees are polite and courteous.					
PSQ8	I feel secure when using MobiFone services.					
PSQ9	MobiFone provides services suited to my individual needs.					
PSQ10	MobiFone transaction points have modern facilities.					
2. Perceived Value (PV)						
PV1	MobiFone's tariffs are reasonable compared with the quality of service I receive.					
PV2	MobiFone's service packages provide many benefits to me.					
PV3	I feel that the money I spend on MobiFone services is worthwhile.					

PV4	MobiFone provides services with competitive value compared with other operators.					

3. Customer Care Quality (CCQ)

CCQ1	MobiFone has many convenient customer-support channels.					
CCQ2	MobiFone resolves complaints quickly.					
CCQ3	I can easily contact MobiFone support staff.					
CCQ4	MobiFone customer-care staff are very helpful.					

4. Perceived Usefulness (PU)

PU1	MobiFone's digital platforms (application, website) help me manage my account effectively.					
PU2	MobiFone's data-network speed (4G/5G) meets my work needs well.					
PU3	MobiFone's data-network speed (4G/5G) meets my entertainment needs well.					
PU4	MobiFone's digital platforms are very useful.					
PU5	MobiFone's digital platforms provide many practical benefits to my life.					

5. Perceived Ease of Use (PEOU)

PEOU1	MobiFone's digital platforms (application, website) are easy to use.					
PEOU2	The interfaces of MobiFone's digital platforms are user-friendly.					
PEOU3	The layout and information presentation on MobiFone's digital platforms are clear and easy to understand.					
PEOU4	The registration/use process for new MobiFone services is simple.					
PEOU5	I have no difficulty finding information on MobiFone's digital platforms.					

6. Customer Satisfaction (CS)

CS1	I am completely satisfied with MobiFone's overall service.					
CS2	MobiFone's service meets my expectations.					
CS3	MobiFone's service exceeds my expectations.					
CS4	I am very satisfied with my experience using MobiFone services.					
CS5	I am satisfied with the services MobiFone provides.					

7. Customer Loyalty (CL)

CL1	I will definitely continue using MobiFone services in the future.					
CL2	I will recommend MobiFone to friends, relatives or colleagues.					

CL3	I tend to prioritize MobiFone over other operators.					
CL4	I will not switch to another operator in the near future.					

PART III. Please provide the following personal information clearly:

1. Gender:

☐ 1. Male 2 ☐. Female

2. Your age

☐ 1. From 18 to 25 ☐ 2. From 26 to 35

☐ 3. From 36 to 5 0 ☐ 4.Over 50

3. Your occupation:

☐ 1. Pupil/Student

☐ 2. Office employee

☐ 3. Teacher/Researcher

☐ 4. Worker /general laborer

☐ 5. Business owner

☐ 6. Other:…………………

4. Marital status

☐ 1. Sing l ☐e 2. Married

5. Your average monthly income, including allowances (m*illion VND)*

☐ 1. Under 5 million ☐ 2. From 5 to under 10 million

☐ 3. From 10 to under 20 million ☐ 4. From 20 million or more

6. Your education level

☐ 1. High school or Vocational secondary school

☐ 2. College

☐ 3. University

☐ 4. Postgraduate

☐ 5. Other: ………………

7. Before using MobiFone services, had you ever used services from another operator?

☐ 1. Yes (please specify): ………………

☐ 2. Never used another operator

Thank you sincerely for your cooperation!

APPENDIX 2: SPSS DATA-PROCESSING RESULTS

Verification results file: M[OBIFONE results.xlxs]

Data collected from Google at 50 MobiFone stores: Google Maps review data from 50 MBF branches

1. Results of descriptive statistical analysis of the survey sample by frequency for qualitative variables

Statistics

		Service	Duration	Purpose	Spending	Gender	Age	Occup.	Marital	Income	Educ.	OtherOp
N	Valid	450	450	450	450	450	450	450	450	450	450	450
	Missing	0	0	0	0	0	0	0	0	0	0	0
Mean		2.64	2.53	1.92	2.29	1.50	2.23	3.00	1.38	2.38	2.66	1.29
Sum		1186	1140	864	1030	674	1002	1348	623	1071	1196	582

Service Used

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Voice service (calls, text messages)	82	18.2	18.2	18.2
	Data service (4G/5G)	139	30.9	30.9	49.1
	Digital services (My MobiFone app, digital content)	90	20.0	20.0	69.1
	All of the above	139	30.9	30.9	100.0
	Total	450	100.0	100.0	

Usage Duration

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	From 6 months to under 1 year	109	24.2	24.2	24.2
	From 1 year to under 3 years	109	24.2	24.2	48.4
	From 3 years to under 5 years	115	25.6	25.6	74.0
	5 years or more	117	26.0	26.0	100.0
	Total	450	100.0	100.0	

Purpose of Use

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Personal communication	183	40.7	40.7	40.7
	Work/study	136	30.2	30.2	70.9
	Entertainment (internet access, watching videos, gaming)	115	25.6	25.6	96.4
	Other	16	3.6	3.6	100.0
	Total	450	100.0	100.0	

Average Spending

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Under VND 100,000	117	26.0	26.0	26.0
	From VND 100,000 to under VND 200,000	142	31.6	31.6	57.6
	From VND 200,000 to under VND 500,000	135	30.0	30.0	87.6
	VND 500,000 or more	56	12.4	12.4	100.0
	Total	450	100.0	100.0	

Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	226	50.2	50.2	50.2
	Female	224	49.8	49.8	100.0
	Total	450	100.0	100.0	

Age

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	From 18 to 25	134	29.8	29.8	29.8
	From 26 to 35	130	28.9	28.9	58.7
	From 36 to 50	136	30.2	30.2	88.9
	Over 50	50	11.1	11.1	100.0
	Total	450	100.0	100.0	

Occupation

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Student	104	23.1	23.1	23.1
	Office employee	96	21.3	21.3	44.4
	Teacher/Researcher	71	15.8	15.8	60.2
	Worker/general laborer	79	17.6	17.6	77.8
	Business owner	77	17.1	17.1	94.9
	Other	23	5.1	5.1	100.0
	Total	450	100.0	100.0	

Marital Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Single	277	61.6	61.6	61.6
	Married	173	38.4	38.4	100.0
	Total	450	100.0	100.0	

Income

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Under 5 million	115	25.6	25.6	25.6
	From 5 to under 10 million	124	27.6	27.6	53.1
	From 10 to under 20 million	136	30.2	30.2	83.3
	20 million or more	75	16.7	16.7	100.0
	Total	450	100.0	100.0	

Education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	High school or vocational secondary school	95	21.1	21.1	21.1
	College	83	18.4	18.4	39.6
	University	176	39.1	39.1	78.7
	Postgraduate	73	16.2	16.2	94.9
	Other	23	5.1	5.1	100.0
	Total	450	100.0	100.0	

Used Another Operator

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Used before	318	70.7	70.7	70.7
	Never used	132	29.3	29.3	100.0
	Total	450	100.0	100.0	

2. Results of descriptive statistical analysis of the survey sample by frequency for quantitative variables

Statistics

		PSQ1	PSQ2	PSQ3	PSQ4	PSQ5	PSQ6	PSQ7	PSQ8	PSQ9	PSQ10
N	Valid	450	450	450	450	450	450	450	450	450	450
	Missing	0	0	0	0	0	0	0	0	0	0
Mean		3.70	3.77	3.70	3.66	3.74	3.75	3.72	3.70	3.69	3.72
Sum		1665	1695	1665	1646	1685	1688	1674	1664	1660	1674

PSQ1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	8	1.8	1.8	1.8
	Disagree	43	9.6	9.6	11.3
	Neutral	117	26.0	26.0	37.3
	Agree	190	42.2	42.2	79.6
	Strongly agree	92	20.4	20.4	100.0
	Total	450	100.0	100.0	

PSQ2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	8	1.8	1.8	1.8
	Disagree	41	9.1	9.1	10.9
	Neutral	107	23.8	23.8	34.7
	Agree	186	41.3	41.3	76.0
	Strongly agree	108	24.0	24.0	100.0
	Total	450	100.0	100.0	

PSQ3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	7	1.6	1.6	1.6
	Disagree	48	10.7	10.7	12.2
	Neutral	114	25.3	25.3	37.6
	Agree	185	41.1	41.1	78.7
	Strongly agree	96	21.3	21.3	100.0
	Total	450	100.0	100.0	

PSQ4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	13	2.9	2.9	2.9
	Disagree	41	9.1	9.1	12.0
	Neutral	117	26.0	26.0	38.0
	Agree	195	43.3	43.3	81.3
	Strongly agree	84	18.7	18.7	100.0
	Total	450	100.0	100.0	

PSQ5

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	10	2.2	2.2	2.2
	Disagree	44	9.8	9.8	12.0
	Neutral	99	22.0	22.0	34.0
	Agree	195	43.3	43.3	77.3
	Strongly agree	102	22.7	22.7	100.0
	Total	450	100.0	100.0	

PSQ6

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	12	2.7	2.7	2.7
	Disagree	39	8.7	8.7	11.3
	Neutral	99	22.0	22.0	33.3
	Agree	199	44.2	44.2	77.6
	Strongly agree	101	22.4	22.4	100.0
	Total	450	100.0	100.0	

PSQ7

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	11	2.4	2.4	2.4
	Disagree	45	10.0	10.0	12.4
	Neutral	102	22.7	22.7	35.1
	Agree	193	42.9	42.9	78.0
	Strongly agree	99	22.0	22.0	100.0
	Total	450	100.0	100.0	

PSQ8

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	11	2.4	2.4	2.4
	Disagree	39	8.7	8.7	11.1
	Neutral	119	26.4	26.4	37.6
	Agree	187	41.6	41.6	79.1
	Strongly agree	94	20.9	20.9	100.0
	Total	450	100.0	100.0	

PSQ9

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	8	1.8	1.8	1.8
	Disagree	45	10.0	10.0	11.8
	Neutral	115	25.6	25.6	37.3
	Agree	193	42.9	42.9	80.2
	Strongly agree	89	19.8	19.8	100.0
	Total	450	100.0	100.0	

PSQ10

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	8	1.8	1.8	1.8
	Disagree	35	7.8	7.8	9.6
	Neutral	124	27.6	27.6	37.1
	Agree	191	42.4	42.4	79.6
	Strongly agree	92	20.4	20.4	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
PSQ1	450	1665	3.70	.958
PSQ2	450	1695	3.77	.975
PSQ3	450	1665	3.70	.972
PSQ4	450	1646	3.66	.978
PSQ5	450	1685	3.74	.987
PSQ6	450	1688	3.75	.986
PSQ7	450	1674	3.72	.995
PSQ8	450	1664	3.70	.975
PSQ9	450	1660	3.69	.958
PSQ10	450	1674	3.72	.935
PSQ	450	1671.60	3.7147	.78167
Valid N (listwise)	450			

Statistics

		PV1	PV2	PV3	PV4
N	Valid	450	450	450	450
	Missing	0	0	0	0
Mean		3.68	3.73	3.71	3.66
Sum		1658	1679	1669	1646

PV1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	11	2.4	2.4	2.4
	Disagree	42	9.3	9.3	11.8
	Neutral	107	23.8	23.8	35.6
	Agree	208	46.2	46.2	81.8
	Strongly agree	82	18.2	18.2	100.0
	Total	450	100.0	100.0	

PV2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	13	2.9	2.9	2.9
	Disagree	35	7.8	7.8	10.7
	Neutral	104	23.1	23.1	33.8
	Agree	206	45.8	45.8	79.6
	Strongly agree	92	20.4	20.4	100.0
	Total	450	100.0	100.0	

PV3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	12	2.7	2.7	2.7
	Disagree	43	9.6	9.6	12.2
	Neutral	95	21.1	21.1	33.3
	Agree	214	47.6	47.6	80.9
	Strongly agree	86	19.1	19.1	100.0
	Total	450	100.0	100.0	

PV4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	11	2.4	2.4	2.4
	Disagree	37	8.2	8.2	10.7
	Neutral	130	28.9	28.9	39.6
	Agree	189	42.0	42.0	81.6
	Strongly agree	83	18.4	18.4	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
PV1	450	1658	3.68	.957
PV2	450	1679	3.73	.968
PV3	450	1669	3.71	.970
PV4	450	1646	3.66	.952
PV	450	1663.00	3.6956	.79554
Valid N (listwise)	450			

Statistics

		CCQ1	CCQ2	CCQ3	CCQ4
N	Valid	450	450	450	450
	Missing	0	0	0	0
Mean		3.93	3.88	3.93	3.90
Sum		1767	1745	1768	1757

CCQ1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	4	.9	.9	.9
	Disagree	40	8.9	8.9	9.8
	Neutral	81	18.0	18.0	27.8
	Agree	185	41.1	41.1	68.9
	Strongly agree	140	31.1	31.1	100.0
	Total	450	100.0	100.0	

CCQ2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	7	1.6	1.6	1.6
	Disagree	30	6.7	6.7	8.2
	Neutral	104	23.1	23.1	31.3
	Agree	179	39.8	39.8	71.1
	Strongly agree	130	28.9	28.9	100.0
	Total	450	100.0	100.0	

CCQ3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	7	1.6	1.6	1.6
	Disagree	35	7.8	7.8	9.3
	Neutral	92	20.4	20.4	29.8
	Agree	165	36.7	36.7	66.4
	Strongly agree	151	33.6	33.6	100.0
	Total	450	100.0	100.0	

CCQ4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	9	2.0	2.0	2.0
	Disagree	24	5.3	5.3	7.3
	Neutral	97	21.6	21.6	28.9
	Agree	191	42.4	42.4	71.3
	Strongly agree	129	28.7	28.7	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
CCQ1	450	1767	3.93	.961
CCQ2	450	1745	3.88	.956
CCQ3	450	1768	3.93	.994
CCQ4	450	1757	3.90	.943
CCQ	450	1759.25	3.9094	.78473
Valid N (listwise)	450			

Statistics

		PU1	PU2	PU3	PU4	PU5
N	Valid	450	450	450	450	450
	Missing	0	0	0	0	0
Mean		3.81	3.80	3.79	3.74	3.81
Sum		1713	1709	1704	1681	1716

PU1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	6	1.3	1.3	1.3
	Disagree	30	6.7	6.7	8.0
	Neutral	116	25.8	25.8	33.8
	Agree	191	42.4	42.4	76.2
	Strongly agree	107	23.8	23.8	100.0
	Total	450	100.0	100.0	

PU2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	6	1.3	1.3	1.3
	Disagree	35	7.8	7.8	9.1
	Neutral	104	23.1	23.1	32.2
	Agree	204	45.3	45.3	77.6
	Strongly agree	101	22.4	22.4	100.0
	Total	450	100.0	100.0	

PU3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	7	1.6	1.6	1.6
	Disagree	29	6.4	6.4	8.0
	Neutral	110	24.4	24.4	32.4
	Agree	211	46.9	46.9	79.3
	Strongly agree	93	20.7	20.7	100.0
	Total	450	100.0	100.0	

PU4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	7	1.6	1.6	1.6
	Disagree	34	7.6	7.6	9.1
	Neutral	113	25.1	25.1	34.2
	Agree	213	47.3	47.3	81.6
	Strongly agree	83	18.4	18.4	100.0
	Total	450	100.0	100.0	

PU5

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	5	1.1	1.1	1.1
	Disagree	29	6.4	6.4	7.6
	Neutral	107	23.8	23.8	31.3
	Agree	213	47.3	47.3	78.7
	Strongly agree	96	21.3	21.3	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
PU1	450	1713	3.81	.920
PU2	450	1709	3.80	.921
PU3	450	1704	3.79	.897
PU4	450	1681	3.74	.900
PU5	450	1716	3.81	.881
PU	450	1704.60	3.7880	.72286
Valid N (listwise)	450			

Statistics

		PEOU1	PEOU2	PEOU3	PEOU4	PEOU5
N	Valid	450	450	450	450	450
	Missing	0	0	0	0	0
Mean		3.78	3.71	3.74	3.73	3.74
Sum		1701	1669	1684	1678	1682

PEOU1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	9	2.0	2.0	2.0
	Disagree	28	6.2	6.2	8.2
	Neutral	113	25.1	25.1	33.3
	Agree	203	45.1	45.1	78.4
	Strongly agree	97	21.6	21.6	100.0
	Total	450	100.0	100.0	

PEOU2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	13	2.9	2.9	2.9
	Disagree	34	7.6	7.6	10.4
	Neutral	117	26.0	26.0	36.4
	Agree	193	42.9	42.9	79.3
	Strongly agree	93	20.7	20.7	100.0
	Total	450	100.0	100.0	

PEOU3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	11	2.4	2.4	2.4
	Disagree	37	8.2	8.2	10.7
	Neutral	100	22.2	22.2	32.9
	Agree	211	46.9	46.9	79.8
	Strongly agree	91	20.2	20.2	100.0
	Total	450	100.0	100.0	

PEOU4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	10	2.2	2.2	2.2
	Disagree	37	8.2	8.2	10.4
	Neutral	115	25.6	25.6	36.0
	Agree	191	42.4	42.4	78.4
	Strongly agree	97	21.6	21.6	100.0
	Total	450	100.0	100.0	

PEOU5

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	9	2.0	2.0	2.0
	Disagree	41	9.1	9.1	11.1
	Neutral	98	21.8	21.8	32.9
	Agree	213	47.3	47.3	80.2
	Strongly agree	89	19.8	19.8	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
PEOU1	450	1701	3.78	.921
PEOU2	450	1669	3.71	.973
PEOU3	450	1684	3.74	.953
PEOU4	450	1678	3.73	.964
PEOU5	450	1682	3.74	.945
PEOU	450	1682.80	3.7396	.78680
Valid N (listwise)	450			

Statistics

		CS1	CS2	CS3	CS4	CS5
N	Valid	450	450	450	450	450
	Missing	0	0	0	0	0
Mean		4.18	4.23	4.15	4.19	4.19
Sum		1883	1903	1866	1884	1887

CS1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	4	.9	.9	.9
	Disagree	16	3.6	3.6	4.4
	Neutral	71	15.8	15.8	20.2
	Agree	161	35.8	35.8	56.0
	Strongly agree	198	44.0	44.0	100.0
	Total	450	100.0	100.0	

CS2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	1	.2	.2	.2
	Disagree	23	5.1	5.1	5.3
	Neutral	59	13.1	13.1	18.4
	Agree	156	34.7	34.7	53.1
	Strongly agree	211	46.9	46.9	100.0
	Total	450	100.0	100.0	

CS3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	5	1.1	1.1	1.1
	Disagree	27	6.0	6.0	7.1
	Neutral	62	13.8	13.8	20.9
	Agree	159	35.3	35.3	56.2
	Strongly agree	197	43.8	43.8	100.0
	Total	450	100.0	100.0	

CS4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	3	.7	.7	.7
	Disagree	22	4.9	4.9	5.6
	Neutral	75	16.7	16.7	22.2
	Agree	138	30.7	30.7	52.9
	Strongly agree	212	47.1	47.1	100.0
	Total	450	100.0	100.0	

CS5

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	5	1.1	1.1	1.1
	Disagree	18	4.0	4.0	5.1
	Neutral	69	15.3	15.3	20.4
	Agree	151	33.6	33.6	54.0
	Strongly agree	207	46.0	46.0	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
CS1	450	1883	4.18	.888
CS2	450	1903	4.23	.880
CS3	450	1866	4.15	.947
CS4	450	1884	4.19	.928
CS5	450	1887	4.19	.915
CS	450	1884.60	4.1880	.66777
Valid N (listwise)	450			

Statistics

		CL1	CL2	CL3	CL4
N	Valid	450	450	450	450
	Missing	0	0	0	0
Mean		4.16	4.15	4.17	4.16
Sum		1871	1867	1875	1872

CL1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	2	.4	.4	.4
	Disagree	17	3.8	3.8	4.2
	Neutral	65	14.4	14.4	18.7
	Agree	190	42.2	42.2	60.9
	Strongly agree	176	39.1	39.1	100.0
	Total	450	100.0	100.0	

CL2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	1	.2	.2	.2
	Disagree	18	4.0	4.0	4.2
	Neutral	77	17.1	17.1	21.3
	Agree	171	38.0	38.0	59.3
	Strongly agree	183	40.7	40.7	100.0
	Total	450	100.0	100.0	

CL3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	2	.4	.4	.4
	Disagree	16	3.6	3.6	4.0
	Neutral	73	16.2	16.2	20.2
	Agree	173	38.4	38.4	58.7
	Strongly agree	186	41.3	41.3	100.0
	Total	450	100.0	100.0	

CL4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Strongly disagree	1	.2	.2	.2
	Disagree	12	2.7	2.7	2.9
	Neutral	77	17.1	17.1	20.0
	Agree	184	40.9	40.9	60.9
	Strongly agree	176	39.1	39.1	100.0
	Total	450	100.0	100.0	

Descriptive Statistics

	N	Sum	Mean	Std. Deviation
CL1	450	1871	4.16	.839
CL2	450	1867	4.15	.859
CL3	450	1875	4.17	.855
CL4	450	1872	4.16	.815
CL	450	1871.25	4.1583	.63541
Valid N (listwise)	450			

3. Cronbach Alpha analysis results

Reliability Statistics

Cronbach's Alpha	N of Items
.939	10

Item Statistics

	Mean	Std. Deviation	N
PSQ1	3.70	.958	450
PSQ2	3.77	.975	450
PSQ3	3.70	.972	450
PSQ4	3.66	.978	450
PSQ5	3.74	.987	450
PSQ6	3.75	.986	450
PSQ7	3.72	.995	450
PSQ8	3.70	.975	450
PSQ9	3.69	.958	450
PSQ10	3.72	.935	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
PSQ1	33.45	49.998	.751	.933
PSQ2	33.38	49.929	.742	.933
PSQ3	33.45	49.954	.742	.933
PSQ4	33.49	49.600	.766	.932
PSQ5	33.40	49.582	.758	.933
PSQ6	33.40	49.634	.756	.933
PSQ7	33.43	49.216	.780	.932
PSQ8	33.45	49.892	.745	.933
PSQ9	33.46	49.737	.773	.932
PSQ10	33.43	50.717	.714	.935

Reliability Statistics

Cronbach's Alpha	N of Items
.846	4

Item Statistics

	Mean	Std. Deviation	N
PV1	3.68	.957	450
PV2	3.73	.968	450
PV3	3.71	.970	450
PV4	3.66	.952	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
PV1	11.10	6.057	.669	.810
PV2	11.05	6.035	.663	.813
PV3	11.07	5.783	.729	.784
PV4	11.12	6.078	.669	.810

Reliability Statistics

Cronbach's Alpha	N of Items
.831	4

Item Statistics

	Mean	Std. Deviation	N
CCQ1	3.93	.961	450
CCQ2	3.88	.956	450
CCQ3	3.93	.994	450
CCQ4	3.90	.943	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
CCQ1	11.71	5.783	.681	.776
CCQ2	11.76	5.978	.634	.797
CCQ3	11.71	5.739	.656	.788
CCQ4	11.73	5.920	.664	.784

Reliability Statistics

Cronbach's Alpha	N of Items
.859	5

Item Statistics

	Mean	Std. Deviation	N
PU1	3.81	.920	450
PU2	3.80	.921	450
PU3	3.79	.897	450
PU4	3.74	.900	450
PU5	3.81	.881	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
PU1	15.13	8.575	.676	.830
PU2	15.14	8.456	.702	.823
PU3	15.15	8.687	.675	.830
PU4	15.20	8.787	.650	.836
PU5	15.13	8.770	.674	.830

Reliability Statistics

Cronbach's Alpha	N of Items
.884	5

Item Statistics

	Mean	Std. Deviation	N
PEOU1	3.78	.921	450
PEOU2	3.71	.973	450
PEOU3	3.74	.953	450
PEOU4	3.73	.964	450
PEOU5	3.74	.945	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
PEOU1	14.92	10.205	.751	.853
PEOU2	14.99	10.127	.711	.862
PEOU3	14.96	10.368	.684	.868
PEOU4	14.97	10.048	.737	.856
PEOU5	14.96	10.208	.725	.859

Reliability Statistics

Cronbach's Alpha	N of Items
.784	5

Item Statistics

	Mean	Std. Deviation	N
CS1	4.18	.888	450
CS2	4.23	.880	450
CS3	4.15	.947	450
CS4	4.19	.928	450
CS5	4.19	.915	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
CS1	16.76	7.668	.548	.748
CS2	16.71	7.591	.574	.739
CS3	16.79	7.291	.579	.737
CS4	16.75	7.456	.559	.744
CS5	16.75	7.593	.539	.751

Reliability Statistics

Cronbach's Alpha	N of Items
.748	4

Item Statistics

	Mean	Std. Deviation	N
CL1	4.16	.839	450
CL2	4.15	.859	450
CL3	4.17	.855	450
CL4	4.16	.815	450

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
CL1	12.48	3.934	.548	.686
CL2	12.48	3.814	.569	.674
CL3	12.47	4.000	.505	.710
CL4	12.47	4.009	.547	.687

4. EFA factor analysis results

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.896
Bartlett's Test of Sphericity	Approx. Chi-Square	6596.854
	df	378
	Sig.	.000

Communalities

	Initial	Extraction
PSQ1	1.000	.646
PSQ2	1.000	.636
PSQ3	1.000	.639
PSQ4	1.000	.666
PSQ5	1.000	.656
PSQ6	1.000	.656
PSQ7	1.000	.685
PSQ8	1.000	.638
PSQ9	1.000	.678
PSQ10	1.000	.598
PV1	1.000	.674
PV2	1.000	.667
PV3	1.000	.742
PV4	1.000	.671
CCQ1	1.000	.698
CCQ2	1.000	.645
CCQ3	1.000	.664
CCQ4	1.000	.675
PU1	1.000	.644
PU2	1.000	.675
PU3	1.000	.644
PU4	1.000	.607
PU5	1.000	.641
PEOU1	1.000	.726
PEOU2	1.000	.675
PEOU3	1.000	.640
PEOU4	1.000	.712
PEOU5	1.000	.688

Extraction Method: Principal Component Analysis.

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	6.490	23.178	23.178	6.490	23.178	23.178	6.488	23.171	23.171
2	3.564	12.728	35.906	3.564	12.728	35.906	3.451	12.326	35.497
3	3.212	11.470	47.376	3.212	11.470	47.376	3.215	11.482	46.979
4	2.792	9.972	57.347	2.792	9.972	57.347	2.760	9.856	56.835
5	2.528	9.028	66.376	2.528	9.028	66.376	2.671	9.540	66.376
6	.601	2.147	68.522						
7	.585	2.088	70.610						
8	.547	1.954	72.564						
9	.542	1.936	74.500						
10	.506	1.805	76.305						
11	.491	1.752	78.057						
12	.479	1.710	79.767						
13	.469	1.674	81.441						
14	.448	1.600	83.041						
15	.437	1.562	84.603						
16	.414	1.477	86.081						
17	.401	1.431	87.512						
18	.394	1.408	88.920						
19	.370	1.323	90.243						
20	.357	1.276	91.519						
21	.345	1.233	92.752						
22	.333	1.191	93.943						
23	.321	1.147	95.091						
24	.300	1.072	96.162						
25	.286	1.020	97.182						
26	.278	.992	98.174						
27	.265	.947	99.122						
28	.246	.878	100.000						

Extraction Method: Principal Component Analysis.

Rotated Component Matrix[a]

	Component 1	2	3	4	5
PSQ7	.828	.013	.005	-.011	-.011
PSQ9	.822	-.021	.038	.035	.019
PSQ4	.815	.000	-.029	-.034	.003
PSQ5	.809	.001	-.015	.020	-.005
PSQ6	.806	.008	-.047	.010	-.063
PSQ1	.802	.046	-.015	.020	-.016
PSQ8	.797	-.040	.006	-.033	.013
PSQ3	.795	-.072	.031	.013	.026
PSQ2	.795	.029	-.007	.058	.001
PSQ10	.770	.038	-.018	-.054	-.004
PEOU1	.032	.850	.001	-.024	.052
PEOU4	-.046	.839	.059	.023	.046
PEOU5	-.010	.828	-.012	.037	-.001
PEOU2	.061	.817	.056	.007	-.025
PEOU3	-.032	.797	-.033	.043	.001
PU2	.025	-.018	.819	.053	.017
PU1	-.060	-.006	.800	.010	-.005
PU3	-.040	.053	.799	.002	-.018
PU5	.030	.045	.798	.025	-.012
PU4	.006	-.008	.778	.035	.001
PV3	.036	.001	.024	.857	.072
PV4	-.016	.048	.008	.815	.066
PV2	-.053	.015	.006	.815	-.005
PV1	.052	.016	.084	.814	-.040
CCQ1	.014	-.008	.003	-.053	.834
CCQ4	-.027	.065	-.027	-.016	.818
CCQ3	.006	.081	.000	.074	.807
CCQ2	-.016	-.072	.006	.083	.795

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 4 iterations.

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.888
Bartlett's Test of Sphericity	Approx. Chi-Square	1111.271
	df	36
	Sig.	<.001

Communalities

	Initial	Extraction
CS1	1.000	.530
CS2	1.000	.554
CS3	1.000	.568
CS4	1.000	.572
CS5	1.000	.492
CL1	1.000	.552
CL2	1.000	.630
CL3	1.000	.496
CL4	1.000	.603

Extraction Method: Principal Component Analysis.

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	3.877	43.078	43.078	3.877	43.078	43.078	2.626	29.174	29.174
2	1.119	12.439	55.517	1.119	12.439	55.517	2.371	26.343	55.517
3	.673	7.480	62.997						
4	.653	7.257	70.254						
5	.584	6.493	76.748						
6	.574	6.378	83.125						
7	.550	6.112	89.237						
8	.509	5.652	94.890						
9	.460	5.110	100.000						

Extraction Method: Principal Component Analysis.

Rotated Component Matrix[a]

	Component 1	Component 2
CS4	.743	.141
CS3	.732	.180
CS2	.704	.240
CS1	.701	.197
CS5	.592	.376
CL2	.161	.777
CL4	.150	.762
CL1	.281	.688
CL3	.268	.651

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 3 iterations.

5. Multiple and simple regression analysis results

Correlations

		CS	PSQ	PV	CCQ	PU	PEOU
Pearson Correlation	CS	1.000	.385	.346	.357	.226	.294
	PSQ	.385	1.000	.008	-.011	-.016	.001
	PV	.346	.008	1.000	.056	.069	.046
	CCQ	.357	-.011	.056	1.000	-.008	.039
	PU	.226	-.016	.069	-.008	1.000	.034
	PEOU	.294	.001	.046	.039	.034	1.000
Sig. (1-tailed)	CS	.	<.001	<.001	<.001	<.001	<.001
	PSQ	.000	.	.430	.405	.367	.491
	PV	.000	.430	.	.119	.073	.167
	CCQ	.000	.405	.119	.	.429	.207
	PU	.000	.367	.073	.429	.	.237
	PEOU	.000	.491	.167	.207	.237	.
N	CS	450	450	450	450	450	450
	PSQ	450	450	450	450	450	450
	PV	450	450	450	450	450	450
	CCQ	450	450	450	450	450	450
	PU	450	450	450	450	450	450
	PEOU	450	450	450	450	450	450

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.704[a]	.496	.490	.47669	2.166

a. Predictors: (Constant), PEOU, PSQ, PU, CCQ, PV

b. Dependent Variable: CS

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	99.323	5	19.865	87.419	<.001[b]
	Residual	100.892	444	.227		
	Total	200.215	449			

a. Dependent Variable: CS

b. Predictors: (Constant), PEOU, PSQ, PU, CCQ, PV

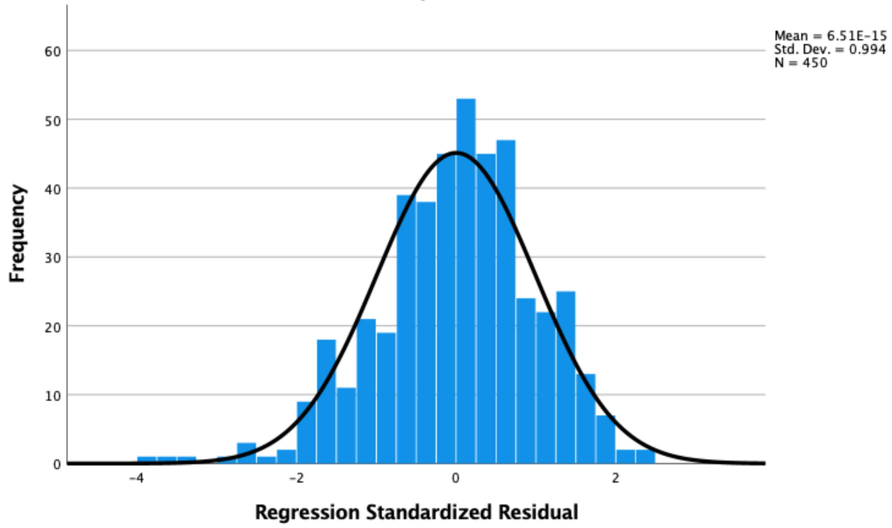
Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	-.638	.237		-2.688	.007		
	PSQ	.333	.029	.389	11.554	<.001	1.000	1.000
	PV	.250	.028	.298	8.807	<.001	.990	1.010
	CCQ	.286	.029	.336	9.955	<.001	.995	1.005
	PU	.190	.031	.206	6.102	<.001	.994	1.006
	PEOU	.221	.029	.260	7.703	<.001	.996	1.004

a. Dependent Variable: CS

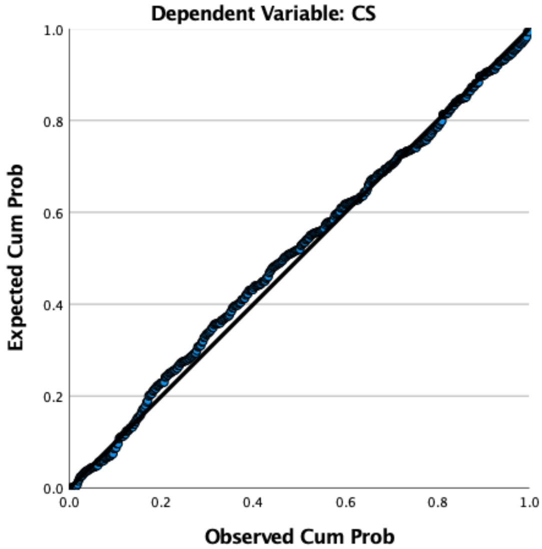
Histogram

Dependent Variable: CS



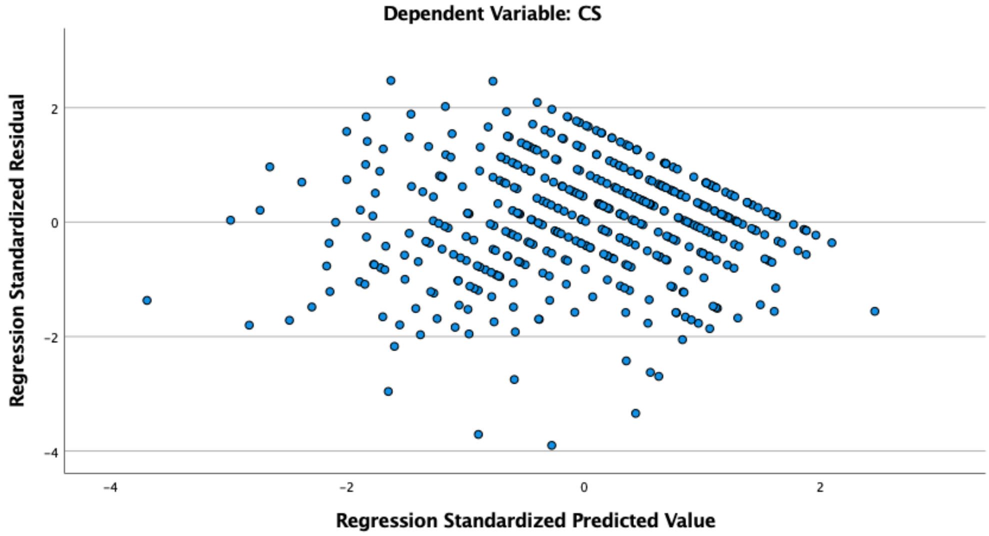
Normal P-P Plot of Regression Standardized Residual

Dependent Variable: CS



Scatterplot

Dependent Variable: CS



Correlations

		CL	CS
Pearson Correlation	CL	1.000	.557
	CS	.557	1.000
Sig. (1-tailed)	CL	.	<.001
	CS	.000	.
N	CL	450	450
	CS	450	450

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.557[a]	.311	.309	.52810	1.951

a. Predictors: (Constant), CS

b. Dependent Variable: CL

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	56.337	1	56.337	202.001	<.001[b]
	Residual	124.944	448	.279		
	Total	181.281	449			

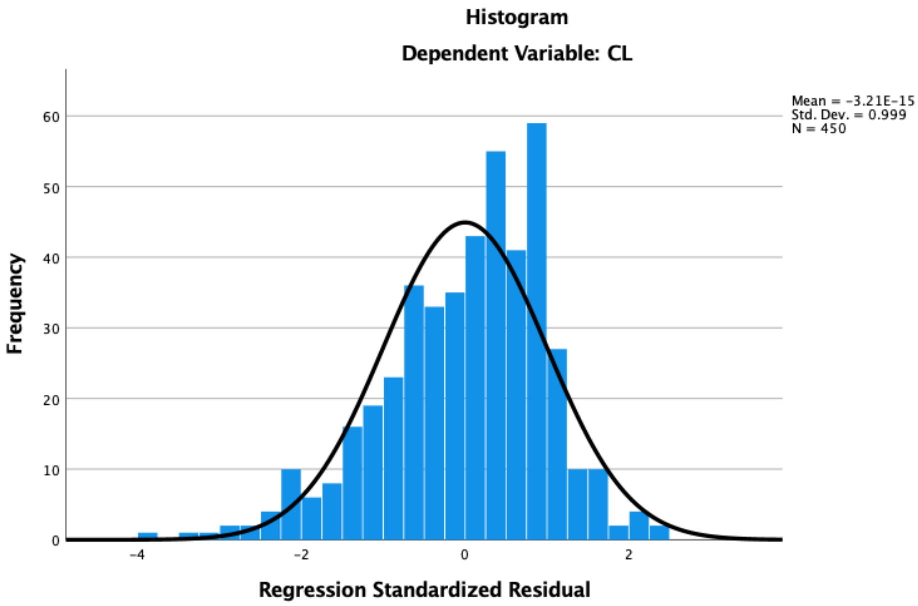
a. Dependent Variable: CL

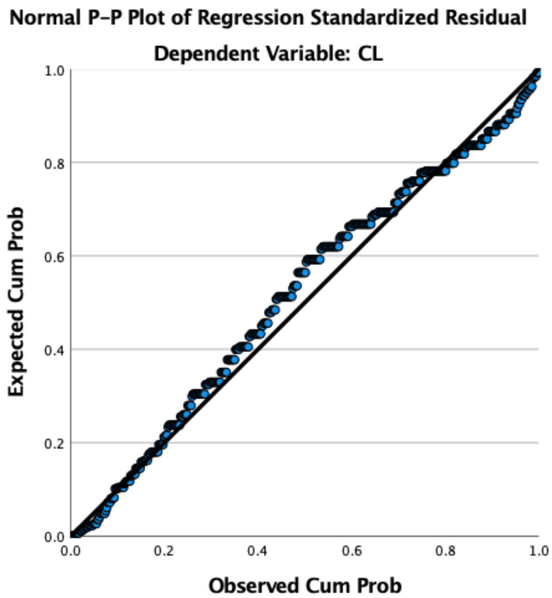
b. Predictors: (Constant), CS

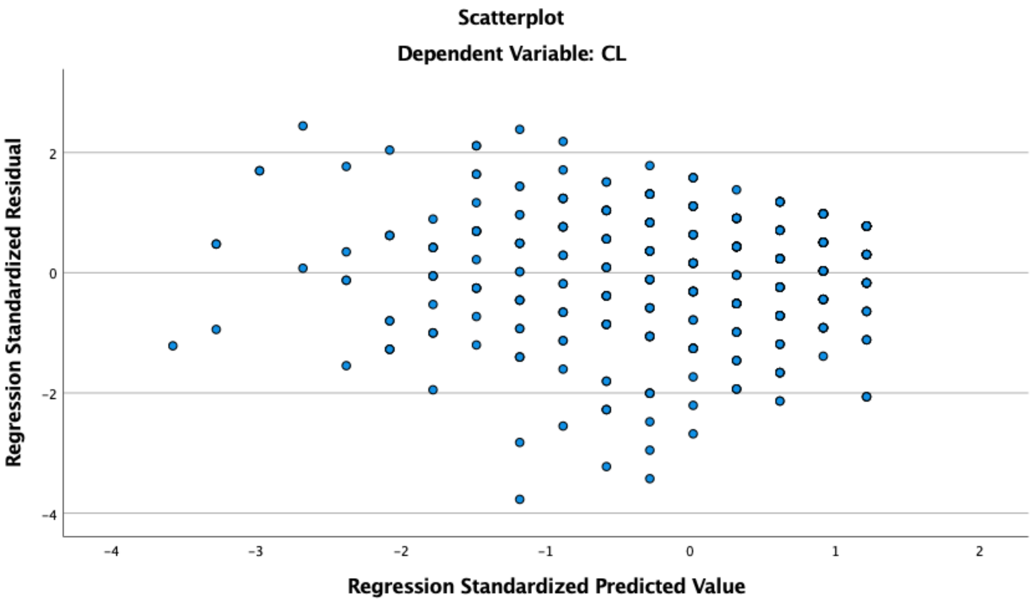
Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	1.937	.158		12.237	<.001		
	CS	.530	.037	.557	14.213	<.001	1.000	1.000

a. Dependent Variable: CL







6. One-sample T-Test results

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
PSQ1	450	3.70	.958	.045
PSQ2	450	3.77	.975	.046
PSQ3	450	3.70	.972	.046
PSQ4	450	3.66	.978	.046
PSQ5	450	3.74	.987	.047
PSQ6	450	3.75	.986	.046
PSQ7	450	3.72	.995	.047
PSQ8	450	3.70	.975	.046
PSQ9	450	3.69	.958	.045
PSQ10	450	3.72	.935	.044
PSQ	450	3.7147	.78167	.03685
PV1	450	3.68	.957	.045
PV2	450	3.73	.968	.046
PV3	450	3.71	.970	.046
PV4	450	3.66	.952	.045
PV	450	3.6956	.79554	.03750
CCQ1	450	3.93	.961	.045
CCQ2	450	3.88	.956	.045
CCQ3	450	3.93	.994	.047
CCQ4	450	3.90	.943	.044
CCQ	450	3.9094	.78473	.03699
PU1	450	3.81	.920	.043
PU2	450	3.80	.921	.043
PU3	450	3.79	.897	.042
PU4	450	3.74	.900	.042
PU5	450	3.81	.881	.042
PU	450	3.7880	.72286	.03408
PEOU1	450	3.78	.921	.043
PEOU2	450	3.71	.973	.046
PEOU3	450	3.74	.953	.045
PEOU4	450	3.73	.964	.045
PEOU5	450	3.74	.945	.045
PEOU	450	3.7396	.78680	.03709
CS1	450	4.18	.888	.042
CS2	450	4.23	.880	.041
CS3	450	4.15	.947	.045
CS4	450	4.19	.928	.044
CS5	450	4.19	.915	.043
CS	450	4.1880	.66777	.03148
CL1	450	4.16	.839	.040
CL2	450	4.15	.859	.040
CL3	450	4.17	.855	.040
CL4	450	4.16	.815	.038
CL	450	4.1583	.63541	.02995

One-Sample Test

	Test Value = 4					
					95% Confidence Interval of the Difference	
	t	df	Sig. (2-tailed)	Mean Difference	Lower	Upper
PSQ1	-6.640	449	<.001	-.300	-.39	-.21
PSQ2	-5.079	449	<.001	-.233	-.32	-.14
PSQ3	-6.545	449	<.001	-.300	-.39	-.21
PSQ4	-7.424	449	<.001	-.342	-.43	-.25
PSQ5	-5.491	449	<.001	-.256	-.35	-.16
PSQ6	-5.357	449	<.001	-.249	-.34	-.16
PSQ7	-5.968	449	<.001	-.280	-.37	-.19
PSQ8	-6.575	449	<.001	-.302	-.39	-.21
PSQ9	-6.886	449	<.001	-.311	-.40	-.22
PSQ10	-6.351	449	<.001	-.280	-.37	-.19
PSQ	-7.743	449	<.001	-.28533	-.3577	-.2129
PV1	-6.995	449	<.001	-.316	-.40	-.23
PV2	-5.894	449	<.001	-.269	-.36	-.18
PV3	-6.363	449	<.001	-.291	-.38	-.20
PV4	-7.622	449	<.001	-.342	-.43	-.25
PV	-8.118	449	<.001	-.30444	-.3781	-.2307
CCQ1	-1.619	449	.106	-.073	-.16	.02
CCQ2	-2.712	449	.007	-.122	-.21	-.03
CCQ3	-1.517	449	.130	-.071	-.16	.02
CCQ4	-2.151	449	.032	-.096	-.18	-.01
CCQ	-2.448	449	.015	-.09056	-.1633	-.0179
PU1	-4.457	449	<.001	-.193	-.28	-.11
PU2	-4.659	449	<.001	-.202	-.29	-.12
PU3	-5.043	449	<.001	-.213	-.30	-.13
PU4	-6.234	449	<.001	-.264	-.35	-.18
PU5	-4.496	449	<.001	-.187	-.27	-.11
PU	-6.221	449	<.001	-.21200	-.2790	-.1450
PEOU1	-5.065	449	<.001	-.220	-.31	-.13
PEOU2	-6.348	449	<.001	-.291	-.38	-.20
PEOU3	-5.736	449	<.001	-.258	-.35	-.17
PEOU4	-5.968	449	<.001	-.271	-.36	-.18
PEOU5	-5.886	449	<.001	-.262	-.35	-.17
PEOU	-7.022	449	<.001	-.26044	-.3333	-.1876
CS1	4.408	449	<.001	.184	.10	.27
CS2	5.520	449	<.001	.229	.15	.31
CS3	3.287	449	.001	.147	.06	.23
CS4	4.269	449	<.001	.187	.10	.27
CS5	4.481	449	<.001	.193	.11	.28
CS	5.972	449	<.001	.18800	.1261	.2499
CL1	3.991	449	<.001	.158	.08	.24
CL2	3.678	449	<.001	.149	.07	.23
CL3	4.133	449	<.001	.167	.09	.25
CL4	4.163	449	<.001	.160	.08	.24
CL	5.286	449	<.001	.15833	.0995	.2172

7. Independent T-Test results

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
PSQ	Male	226	3.7398	.80714	.05369
	Female	224	3.6893	.75606	.05052
PV	Male	226	3.6726	.75578	.05027
	Female	224	3.7188	.83480	.05578
CCQ	Male	226	3.7500	.74125	.04931
	Female	224	4.0703	.79609	.05319
PU	Male	226	3.7973	.72455	.04820
	Female	224	3.7786	.72265	.04828
PEOU	Male	226	3.7425	.77635	.05164
	Female	224	3.7366	.79893	.05338
CS	Male	226	4.1027	.68643	.04566
	Female	224	4.2741	.63842	.04266
CL	Male	226	4.1648	.66139	.04399
	Female	224	4.1518	.60949	.04072

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means					95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
PSQ	Equal variances assumed	.751	.387	.685	448	.493	.05054	.07374	-.09438	.19546
	Equal variances not assumed			.686	446.580	.493	.05054	.07372	-.09434	.19542
PV	Equal variances assumed	2.104	.148	-.615	448	.539	-.04618	.07506	-.19369	.10132
	Equal variances not assumed			-.615	442.837	.539	-.04618	.07509	-.19376	.10139
CCQ	Equal variances assumed	2.197	.139	-4.418	448	<.001	-.32031	.07251	-.46281	-.17782
	Equal variances not assumed			-4.416	445.140	<.001	-.32031	.07253	-.46285	-.17777
PU	Equal variances assumed	.013	.909	.275	448	.783	.01877	.06822	-.11530	.15285
	Equal variances not assumed			.275	447.982	.783	.01877	.06822	-.11530	.15285
PEOU	Equal variances assumed	.035	.851	.079	448	.937	.00587	.07426	-.14008	.15182
	Equal variances not assumed			.079	447.369	.937	.00587	.07427	-.14010	.15184
CS	Equal variances assumed	2.068	.151	-2.743	448	.006	-.17145	.06251	-.29429	-.04861
	Equal variances not assumed			-2.744	446.201	.006	-.17145	.06249	-.29426	-.04865
CL	Equal variances assumed	.992	.320	.217	448	.828	.01304	.05997	-.10482	.13090
	Equal variances not assumed			.217	445.647	.828	.01304	.05995	-.10478	.13086

8. ANOVA test results

Descriptives

		N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound	95% Confidence Interval for Mean Upper Bound	Minimum	Maximum
PSQ	From 6 months to under 1 year	109	3.7459	.73290	.07020	3.6067	3.8850	1.60	5.00
	From 1 year to under 3 years	109	3.7229	.80974	.07756	3.5692	3.8767	1.70	5.00
	From 3 years to under 5 years	115	3.7313	.82710	.07713	3.5785	3.8841	1.30	5.00
	5 years or more	117	3.6615	.76031	.07029	3.5223	3.8008	1.30	5.00
	Total	450	3.7147	.78167	.03685	3.6423	3.7871	1.30	5.00
PV	From 6 months to under 1 year	109	3.7638	.74910	.07175	3.6215	3.9060	1.00	5.00
	From 1 year to under 3 years	109	3.7431	.76259	.07304	3.5983	3.8879	1.25	5.00
	From 3 years to under 5 years	115	3.7478	.76984	.07179	3.6056	3.8900	1.25	5.00
	5 years or more	117	3.5363	.87670	.08105	3.3758	3.6969	1.00	5.00
	Total	450	3.6956	.79554	.03750	3.6219	3.7693	1.00	5.00
CCQ	From 6 months to under 1 year	109	3.8670	.85430	.08183	3.7048	4.0292	1.50	5.00
	From 1 year to under 3 years	109	3.9450	.74873	.07172	3.8028	4.0871	2.00	5.00
	From 3 years to under 5 years	115	3.9543	.71603	.06677	3.8221	4.0866	1.00	5.00
	5 years or more	117	3.8718	.81976	.07579	3.7217	4.0219	1.25	5.00
	Total	450	3.9094	.78473	.03699	3.8367	3.9821	1.00	5.00
PU	From 6 months to under 1 year	109	3.8183	.78057	.07477	3.6702	3.9665	1.40	5.00
	From 1 year to under 3 years	109	3.7835	.67612	.06476	3.6551	3.9119	1.80	5.00
	From 3 years to under 5 years	115	3.6922	.74800	.06975	3.5540	3.8304	1.20	5.00
	5 years or more	117	3.8581	.68181	.06303	3.7333	3.9830	1.80	5.00
	Total	450	3.7880	.72286	.03408	3.7210	3.8550	1.20	5.00
PEOU	From 6 months to under 1 year	109	3.7028	.73918	.07080	3.5624	3.8431	1.80	5.00
	From 1 year to under 3 years	109	3.8661	.72677	.06961	3.7281	4.0040	1.40	5.00
	From 3 years to under 5 years	115	3.7026	.85157	.07941	3.5453	3.8599	1.40	5.00
	5 years or more	117	3.6923	.81433	.07528	3.5432	3.8414	1.40	5.00
	Total	450	3.7396	.78680	.03709	3.6667	3.8124	1.40	5.00
CS	From 6 months to under 1 year	109	3.9413	.64882	.06215	3.8181	4.0645	2.00	5.00
	From 1 year to under 3 years	109	4.2477	.55304	.05297	4.1427	4.3527	2.40	5.00
	From 3 years to under 5 years	115	4.1826	.70664	.06589	4.0521	4.3131	1.80	5.00
	5 years or more	117	4.3675	.68227	.06308	4.2426	4.4925	2.00	5.00
	Total	450	4.1880	.66777	.03148	4.1261	4.2499	1.80	5.00
CL	From 6 months to under 1 year	109	3.7798	.60066	.05753	3.6658	3.8939	1.75	5.00
	From 1 year to under 3 years	109	4.0665	.61203	.05862	3.9503	4.1827	2.25	5.00
	From 3 years to under 5 years	115	4.3022	.59281	.05528	4.1927	4.4117	2.25	5.00
	5 years or more	117	4.4551	.52994	.04899	4.3581	4.5522	2.50	5.00
	Total	450	4.1583	.63541	.02995	4.0995	4.2172	1.75	5.00

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
PSQ	Based on Mean	.552	3	446	.647
	Based on Median	.515	3	446	.672
	Based on Median and with adjusted df	.515	3	441.881	.672
	Based on trimmed mean	.546	3	446	.651
PV	Based on Mean	1.617	3	446	.185
	Based on Median	1.524	3	446	.207
	Based on Median and with adjusted df	1.524	3	436.593	.208
	Based on trimmed mean	1.596	3	446	.190
CCQ	Based on Mean	2.621	3	446	.050
	Based on Median	1.930	3	446	.124
	Based on Median and with adjusted df	1.930	3	439.180	.124
	Based on trimmed mean	2.467	3	446	.062
PU	Based on Mean	1.393	3	446	.244
	Based on Median	.868	3	446	.458
	Based on Median and with adjusted df	.868	3	429.621	.458
	Based on trimmed mean	1.310	3	446	.270
PEOU	Based on Mean	1.349	3	446	.258
	Based on Median	1.285	3	446	.279
	Based on Median and with adjusted df	1.285	3	434.554	.279
	Based on trimmed mean	1.311	3	446	.270
CS	Based on Mean	1.635	3	446	.180
	Based on Median	1.459	3	446	.225
	Based on Median and with adjusted df	1.459	3	413.832	.225
	Based on trimmed mean	1.457	3	446	.225
CL	Based on Mean	.886	3	446	.448
	Based on Median	.956	3	446	.413
	Based on Median and with adjusted df	.956	3	430.219	.413
	Based on trimmed mean	1.227	3	446	.299

ANOVA

		Sum of Squares	df	Mean Square	F	Sig.
PSQ	Between Groups	.476	3	.159	.258	.855
	Within Groups	273.868	446	.614		
	Total	274.343	449			
PV	Between Groups	4.034	3	1.345	2.141	.094
	Within Groups	280.132	446	.628		
	Total	284.166	449			
CCQ	Between Groups	.732	3	.244	.395	.757
	Within Groups	275.766	446	.618		
	Total	276.497	449			
PU	Between Groups	1.734	3	.578	1.107	.346
	Within Groups	232.881	446	.522		
	Total	234.615	449			
PEOU	Between Groups	2.310	3	.770	1.246	.293
	Within Groups	275.646	446	.618		
	Total	277.956	449			
CS	Between Groups	10.797	3	3.599	8.474	<.001
	Within Groups	189.418	446	.425		
	Total	200.215	449			
CL	Between Groups	29.221	3	9.740	28.569	<.001
	Within Groups	152.060	446	.341		
	Total	181.281	449			

Multiple Comparisons

Tamhane

Dependent Variable	(I) Usage duration	(J) Usage duration	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
CS	From 6 months to under 1 year	From 1 year to under 3 years	-.30642*	.08166	.001	-.5233	-.0895
		From 3 years to under 5 years	-.24132*	.09058	.049	-.4818	-.0009
		5 years or more	-.42624*	.08855	<.001	-.6613	-.1912
	From 1 year to under 3 years	From 6 months to under 1 year	.30642*	.08166	.001	.0895	.5233
		From 3 years to under 5 years	.06510	.08455	.970	-.1594	.2896
		5 years or more	-.11981	.08237	.615	-.3385	.0989
	From 3 years to under 5 years	From 6 months to under 1 year	.24132*	.09058	.049	.0009	.4818
		From 1 year to under 3 years	-.06510	.08455	.970	-.2896	.1594
		5 years or more	-.18491	.09122	.236	-.4270	.0572
	5 years or more	From 6 months to under 1 year	.42624*	.08855	<.001	.1912	.6613
		From 1 year to under 3 years	.11981	.08237	.615	-.0989	.3385
		From 3 years to under 5 years	.18491	.09122	.236	-.0572	.4270
CL	From 6 months to under 1 year	From 1 year to under 3 years	-.28670*	.08214	.004	-.5048	-.0686
		From 3 years to under 5 years	-.52236*	.07979	<.001	-.7342	-.3105
		5 years or more	-.67531*	.07557	<.001	-.8760	-.4747
	From 1 year to under 3 years	From 6 months to under 1 year	.28670*	.08214	.004	.0686	.5048
		From 3 years to under 5 years	-.23566*	.08058	.023	-.4496	-.0217
		5 years or more	-.38861*	.07640	<.001	-.5915	-.1857
	From 3 years to under 5 years	From 6 months to under 1 year	.52236*	.07979	<.001	.3105	.7342
		From 1 year to under 3 years	.23566*	.08058	.023	.0217	.4496
		5 years or more	-.15295	.07387	.215	-.3490	.0431
	5 years or more	From 6 months to under 1 year	.67531*	.07557	<.001	.4747	.8760
		From 1 year to under 3 years	.38861*	.07640	<.001	.1857	.5915
		From 3 years to under 5 years	.15295	.07387	.215	-.0431	.3490

*. The mean difference is significant at the 0.05 level.

Multiple Comparisons

Tamhane

Dependent Variable	(I) Usage duration	(J) Usage duration	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
CS	From 6 months to under 1 year	From 1 year to under 3 years	-.30642*	.08166	.001	-.5233	-.0895
		From 3 years to under 5 years	-.24132*	.09058	.049	-.4818	-.0009
		5 years or more	-.42624*	.08855	<.001	-.6613	-.1912
	From 1 year to under 3 years	From 6 months to under 1 year	.30642*	.08166	.001	.0895	.5233
		From 3 years to under 5 years	.06510	.08455	.970	-.1594	.2896
		5 years or more	-.11981	.08237	.615	-.3385	.0989
	From 3 years to under 5 years	From 6 months to under 1 year	.24132*	.09058	.049	.0009	.4818
		From 1 year to under 3 years	-.06510	.08455	.970	-.2896	.1594
		5 years or more	-.18491	.09122	.236	-.4270	.0572
	5 years or more	From 6 months to under 1 year	.42624*	.08855	<.001	.1912	.6613
		From 1 year to under 3 years	.11981	.08237	.615	-.0989	.3385
		From 3 years to under 5 years	.18491	.09122	.236	-.0572	.4270
CL	From 6 months to under 1 year	From 1 year to under 3 years	-.28670*	.08214	.004	-.5048	-.0686
		From 3 years to under 5 years	-.52236*	.07979	<.001	-.7342	-.3105
		5 years or more	-.67531*	.07557	<.001	-.8760	-.4747
	From 1 year to under 3 years	From 6 months to under 1 year	.28670*	.08214	.004	.0686	.5048
		From 3 years to under 5 years	-.23566*	.08058	.023	-.4496	-.0217
		5 years or more	-.38861*	.07640	<.001	-.5915	-.1857
	From 3 years to under 5 years	From 6 months to under 1 year	.52236*	.07979	<.001	.3105	.7342
		From 1 year to under 3 years	.23566*	.08058	.023	.0217	.4496
		5 years or more	-.15295	.07387	.215	-.3490	.0431
	5 years or more	From 6 months to under 1 year	.67531*	.07557	<.001	.4747	.8760
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		From 3 years to under 5 years	.15295	.07387	.215	-.0431	.3490

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		From 3 years to under 5 years	-.24132*	.09058	.049	-.4818	-.0009
		5 years or more	-.42624*	.08855	<.001	-.6613	-.1912
	From 1 year to under 3 years	From 6 months to under 1 year	.30642*	.08166	.001	.0895	.5233
		From 3 years to under 5 years	.06510	.08455	.970	-.1594	.2896
		5 years or more	-.11981	.08237	.615	-.3385	.0989
	From 3 years to under 5 years	From 6 months to under 1 year	.24132*	.09058	.049	.0009	.4818
		From 1 year to under 3 years	-.06510	.08455	.970	-.2896	.1594
		5 years or more	-.18491	.09122	.236	-.4270	.0572
	5 years or more	From 6 months to under 1 year	.42624*	.08855	<.001	.1912	.6613
		From 1 year to under 3 years	.11981	.08237	.615	-.0989	.3385
		From 3 years to under 5 years	.18491	.09122	.236	-.0572	.4270
CL	From 6 months to under 1 year	From 1 year to under 3 years	-.28670*	.08214	.004	-.5048	-.0686
		From 3 years to under 5 years	-.52236*	.07979	<.001	-.7342	-.3105
		5 years or more	-.67531*	.07557	<.001	-.8760	-.4747
	From 1 year to under 3 years	From 6 months to under 1 year	.28670*	.08214	.004	.0686	.5048
		From 3 years to under 5 years	-.23566*	.08058	.023	-.4496	-.0217
		5 years or more	-.38861*	.07640	<.001	-.5915	-.1857
	From 3 years to under 5 years	From 6 months to under 1 year	.52236*	.07979	<.001	.3105	.7342
		From 1 year to under 3 years	.23566*	.08058	.023	.0217	.4496
		5 years or more	-.15295	.07387	.215	-.3490	.0431
	5 years or more	From 6 months to under 1 year	.67531*	.07557	<.001	.4747	.8760
		From 1 year to under 3 years	.38861*	.07640	<.001	.1857	.5915
		From 3 years to under 5 years	.15295	.07387	.215	-.0431	.3490

*. The mean difference is significant at the 0.05 level.